Lead Quality Analysis

Debt Reduction Case Study

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1 Phase 1: Data Profiling

The first step before any analysis is to understand exactly what we are working with — dimensions, data types, date range, and a high-level profile of every column.

1.1 Dataset Dimensions

```
##  Data loaded successfully
```

```
## Rows: 3021
```

```
## Columns: 24
```

The dataset contains 3,021 leads recorded across 24 variables, covering a 6-month period from April to September 2009. Each row represents one individual who submitted a debt reduction enquiry online. With over 3,000 records, the dataset provides sufficient volume for statistically meaningful segment-level analysis.

1.2 Data Types and Column Structure

```
## Rows: 3,021
## Columns: 24
## $ LeadCreated                   <dttm> 2009-07-01 01:24:29, 2009-04-13 15:27:3~
## $ FirstName                     <chr> "Dorinda", "Presetta", "Gina", "Kari", "~
## $ Email                         <chr> "kanani@sandwichisles.net", "clerk2@ustc~
## $ VendorLeadID                  <chr> "FDF81FDA-A649-437B-B99C-FDDE74F7971B", ~
## $ CallStatus                    <chr> NA, NA, "Unable to contact - Bad Contact~
## $ WidgetName                    <chr> "w-302252-DebtReduction1-1DC-CreditSolut~
## $ PublisherZoneName             <chr> "TopLeft-302252", "TopLeft-302252", "Top~
## $ PublisherCampaignName         <chr> "DebtReductionInc", "DebtReductionInc", ~
## $ AddressScore                  <chr> "NULL", "NULL", "NULL", "5", "NULL", "5"~
## $ PhoneScore                    <chr> "5", "NULL", "NULL", "3", "NULL", "3", "~
## $ AdvertiserCampaignName        <chr> "creditsolutions-branded-shortform", "De~
## $ State                         <chr> "HI", "AL", "NY", "WA", "TX", "CO", "MI"~
## $ DebtLevel                     <chr> "30001-50000", "90000-100000", "20001-30~
## $ `IP Address`                  <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ Partner                       <chr> "google", "AdKnowledge", "AdKnowledge", ~
## $ ReferralDomain                <chr> "www.google.com", NA, "us.mc582.mail.yah~
## $ MarketingCampaign             <chr> "Debt Holding Tank", "Financial Services~
## $ AdGroup                       <chr> "Holding Tank - Debt", "Consolidate", "C~
## $ Keyword                       <chr> "Debt specialists", NA, NA, NA, NA, NA, ~
## $ SearchQuery                   <chr> "debt specialists", NA, NA, NA, NA, NA, ~
## $ ReferralURL                   <chr> "http://www.google.com/search", NA, "htt~
## $ `ReferralURL Parameters`      <chr> "sourceid=navclient&aq=1&oq=debt sp&ie=U~
## $ LandingPageURL                <chr> "http://www.debtreductioninc.com/index8.~
## $ `Landing Page URL Parameters` <chr> "utm_source=google&utm_medium=CPC&utm_co~
```

R correctly identified LeadCreated as a datetime field, which means no conversion is needed for time-based analysis. However, two notable type issues were detected: AddressScore and PhoneScore are stored as character type rather than numeric, and the IP Address column is entirely empty (logical type with no values). Both will be addressed in the cleaning phase.

1.3 Date Range

Table 1: Dataset Date Range

Earliest_Lead	Latest_Lead	Total_Days
2009-04-02 18:52:41	2009-09-30 21:00:20	181.0886

- Data spans **April 2, 2009 to September 30, 2009** - exactly 181 days (approximately 6 months)
- No dates fall outside this expected window - no data integrity issues detected in the time dimension
- The 6-month window provides sufficient volume for monthly trend analysis across at least 6 data points

1.4 CallStatus Distribution

Table 2: CallStatus Frequency Table

Status	Count	Percentage	Cumulative
NA	2140	70.8	70.8
Closed	245	8.1	78.9
Contacted - Doesn't Qualify	212	7.0	85.9
Unable to contact - Bad Contact Information	191	6.3	92.2
EP Confirmed	130	4.3	96.5
Contacted - Invalid Profile	85	2.8	99.3
EP Sent	16	0.5	99.8
EP Received	2	0.1	99.9

1.5 Full Column Profile

Table 3: Data summary

Name	raw_data
Number of rows	3021
Number of columns	24
Column type frequency:	
character	22
logical	1
POSIXct	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
FirstName	0	1.00	1	17	0	1695	0
Email	0	1.00	9	35	0	2888	0
VendorLeadID	8	1.00	22	36	0	3012	0
CallStatus	2140	0.29	6	43	0	7	0
WidgetName	0	1.00	27	44	0	14	0
PublisherZoneName	0	1.00	14	17	0	2	0
PublisherCampaignName	0	1.00	16	23	0	2	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
AddressScore	0	1.00	1	4	0	6	0
PhoneScore	0	1.00	1	4	0	6	0
AdvertiserCampaignName	0	1.00	23	33	0	2	0
State	0	1.00	2	2	0	32	0
DebtLevel	0	1.00	10	16	0	10	0
Partner	0	1.00	5	13	0	6	0
ReferralDomain	515	0.83	8	35	0	372	0
MarketingCampaign	272	0.91	4	45	0	20	0
AdGroup	272	0.91	3	37	0	103	0
Keyword	2042	0.32	3	48	0	285	0
SearchQuery	1756	0.42	5	85	0	960	0
ReferralURL	515	0.83	20	777	0	670	0
ReferralURL Parameters	738	0.76	10	1365	0	2199	0
LandingPageURL	0	1.00	43	70	0	5	0
Landing Page URL Parameters	0	1.00	16	353	0	1090	0

Variable type: logical

skim_variable	n_missing	complete_rate	mean	count
IP Address	3021	0	NaN	:

Variable type: POSIXct

skim_variable	n_missing	complete_rate	min	max	median	n_unique
LeadCreated	0	1	2009-04-02 18:52:41	2009-09-30 21:00:20	2009-06-18 19:45:45	2786

- **`Keyword`**: 67.6% missing - expected, as not all traffic comes from search keywords
- **`ReferralDomain`**: 17% missing - some leads came directly without a referral
- **`MarketingCampaign` and `AdGroup`**: 9% missing - minor gap, low impact on analysis
- **`VendorLeadID`**: 8 missing values - negligible
- **`WidgetName`**: 14 unique ad designs - will require simplification in Phase 4 Feature Engineering
- **`DebtLevel`**: 10 unique categories - will be converted to an ordered factor in Phase 3

1.6 Unique Value Counts

Table 7: Unique Value Count Per Column

Column	Unique_Values
VendorLeadID	3013
Email	2888
ReferralURL Parameters	2200
FirstName	1695
Landing Page URL Parameters	1090
SearchQuery	961

Column	Unique_Values
ReferralURL	671
ReferralDomain	373
Keyword	286
AdGroup	104
State	32
MarketingCampaign	21
WidgetName	14
DebtLevel	10
CallStatus	8
AddressScore	6
PhoneScore	6
Partner	6
LandingPageURL	5
PublisherZoneName	2
PublisherCampaignName	2
AdvertiserCampaignName	2

- `VendorLeadID` has 3,013 unique values - confirms it is a near-unique identifier, not useful for grouping
- `Partner` has **6 unique values** - expected only 3 (Google, Yahoo, AdKnowledge) - possible capitalisation inconsistency requiring investigation in Phase 2
- `State` = 32 unique values - good for geographic segmentation
- `AdvertiserCampaignName` = 2 values - clean binary split (Branded vs Generic)
- `PublisherCampaignName` = 2 values - clean binary split (Online vs Call Center)

1.7 Deep Investigation

Three issues flagged during profiling require immediate investigation before proceeding to data quality assessment. These findings will directly inform cleaning decisions in Phase 3.

```
## === ALL PARTNER VALUES ===

## # A tibble: 6 x 3
##   Partner           n   pct
##   <chr>         <int> <dbl>
## 1 google          979  32.4
## 2 yahoo           958  31.7
## 3 Google          639  21.2
## 4 Call_Center     271   9
## 5 AdKnowledge     171   5.7
## 6 Advertise.com     3   0.1

##
## === NA CallStatus by Month ===
```

```
## # A tibble: 6 x 4
##   month   total_leads missing_status missing_pct
##   <chr>         <int>          <int>       <dbl>
## 1 2009-04         518            382        73.7
## 2 2009-05         580            464        80
## 3 2009-06         609            382        62.7
## 4 2009-07         532            373        70.1
## 5 2009-08         438            309        70.5
## 6 2009-09         344            230        66.9
```

```
##
## === AddressScore Values ===
```

```
## # A tibble: 6 x 2
##   AddressScore     n
##   <chr>        <int>
## 1 NULL          1850
## 2 5              870
## 3 1              122
## 4 3               93
## 5 4               48
## 6 2               38
```

```
##
## === PhoneScore Values ===
```

```
## # A tibble: 6 x 2
##   PhoneScore     n
##   <chr>      <int>
## 1 NULL        1628
## 2 5            619
## 3 3            427
## 4 4            184
## 5 2            162
## 6 1              1
```

Key Findings:

Partner Column

The Partner column returned 6 distinct values rather than the 3 expected. The breakdown revealed that "google" (979 leads, 32.4%) and "Google" (639 leads, 21.2%) are being treated as two separate partners due to inconsistent capitalisation. Combined, Google accounts for 53.6% of all leads. Two additional sources also appeared: "Call_Center" (271 leads, 9.0%) representing phone-based leads, and "Advertise.com" (3 leads, 0.1%), a negligible source. All capitalisation inconsistencies will be standardised in Phase 3.

CallStatus Missing Values

The 70.8% missing rate in CallStatus was investigated across all six months to determine whether it was isolated to a specific period or systemic across the full dataset. The results confirmed it is systemic, with missing rates ranging from 62.7% in June to 80.0% in May, consistently high throughout the entire data collection window. This pattern strongly suggests these are leads that were never followed up on or whose outcomes were never recorded in the system, rather than a technical data loss event. They will be retained in the dataset and treated as a separate "No Outcome" category rather than being dropped.

AddressScore and PhoneScore

Both score columns were stored as character type containing the string "NULL" rather than proper missing values. AddressScore had 1,850 NULL entries (61.2%) with valid scores ranging from 1 to 5. PhoneScore had 1,628 NULL entries (53.9%) with valid scores also ranging from 1 to 5, with a single entry of score 1. The "NULL" strings will be converted to NA and both columns will be cast to numeric in Phase 3. The valid scale of 1 to 5 is confirmed with no unexpected values present.

2 PHASE 2: DATA QUALITY ASSESSMENT

2.1 Duplicate Check

```
## Duplicate Emails: 133
```

```
## Duplicate VendorLeadIDs: 1
```

133 duplicate email addresses were detected across 3,021 leads, representing 4.4% of total records. These likely reflect the same person submitting multiple enquiries, which inflates raw lead volume but does not represent unique prospects. A single duplicate VendorLeadID was also found, which is negligible. Duplicate emails will be flagged and retained for now but excluded from quality rate calculations in Phase 5 to avoid double-counting.

2.2 Partner Capitalisation Audit

```
## # A tibble: 6 x 3
##   Partner       Partner_clean     n
##   <chr>         <chr>         <int>
## 1 google        Google          979
## 2 yahoo         Yahoo           958
## 3 Google        Google          639
## 4 Call_Center   Call_center     271
## 5 AdKnowledge   Adknowledge     171
## 6 Advertise.com Advertise.com     3
```

The audit confirmed the capitalisation inconsistency flagged in Phase 1. "google" and "Google" are the same partner and will be merged, giving Google a combined total of 1,618 leads (53.6% of all traffic). Note: str_to_title() produces unintended results for two values: "Call_Center" becomes "Call_center" (underscore breaks capitalisation) and "AdKnowledge" becomes "Adknowledge" (loses the capital K). These two will be manually recoded using case_when() in Phase 3 rather than relying on str_to_title() alone.

2.3 State Value Audit

```
## # A tibble: 32 x 2
##    State     n
##    <chr> <int>
##  1 CA      463
##  2 TX      325
##  3 NY      283
##  4 FL      261
##  5 IL      186
##  6 MI      145
##  7 VA      145
```

```
##  8 CO      105
##  9 MD       98
## 10 AL       96
## 11 AZ       94
## 12 MO       90
## 13 LA       80
## 14 PA       75
## 15 MA       70
## 16 NV       54
## 17 OR       54
## 18 WV       52
## 19 OK       50
## 20 AR       48
## 21 IN       36
## 22 HI       32
## 23 NM       29
## 24 WA       28
## 25 MT       24
## 26 CT       23
## 27 NE       22
## 28 IA       19
## 29 SD       11
## 30 ND        9
## 31 AK        8
## 32 DC        6
```

All 32 state values are clean 2-letter US abbreviations with no full names, typos, or unexpected entries. No cleaning is required for this column. The top five states by lead volume are California (463), Texas (325), New York (283), Florida (261), and Illinois (186), which together account for 52.8% of all leads. These states will be primary segments in Phase 5 EDA.

2.4 DebtLevel Ordering Audit

```
## # A tibble: 10 x 2
##    DebtLevel            n
##    <chr>            <int>
##  1 10001-15000        291
##  2 15001-20000        408
##  3 20001-30000        456
##  4 30001-50000        496
##  5 50001-70000        245
##  6 70001-90000        131
##  7 7500-10000         442
##  8 7500-15000         271
##  9 90000-100000        90
## 10 More_than_100000   191
```

DebtLevel contains 10 categories currently stored as unordered character strings. Two issues were identified: first, "7500-10000" (442 leads) and "7500-15000" (271 leads) are overlapping ranges, suggesting the lower bound was captured differently across different form versions. Second, "More_than_100000" uses an underscore and does not sort correctly alphabetically. Both issues will be resolved in Phase 3 by converting DebtLevel to an ordered factor with explicit level definitions.

2.5 AddressScore and PhoneScore NULL Audit

```
## AddressScore NULLs: 1850
```

```
## PhoneScore NULLs: 1628
```

AddressScore contains 1,850 string "NULL" entries (61.2% of rows) and PhoneScore contains 1,628 (53.9%). Both columns store values as character type. Valid scores range from 1 to 5 with no unexpected values outside this scale. The "NULL" strings will be replaced with proper NA values using na_if() and both columns will be cast to numeric in Phase 3. The high missing rate is expected, as the assignment notes these scores were only introduced partway through the data collection period.

2.6 WidgetName Audit

```
## # A tibble: 2 x 2
##   widget_size     n
##   <chr>       <int>
## 1 300250        849
## 2 302252       2172
```

The audit confirmed that widget sizes 300250 (849 leads, 28.1%) and 302252 (2,172 leads, 71.9%) are the only two sizes present. As documented in the assignment brief, these two sizes are functionally identical and represent the same ad format. They will be standardised to a single size label in Phase 4 Feature Engineering to prevent artificial splits during segment analysis.

2.7 CallStatus Values Audit

```
## # A tibble: 8 x 2
##   CallStatus                                       n
##   <chr>                                        <int>
## 1 <NA>                                          2140
## 2 Closed                                         245
## 3 Contacted - Doesn't Qualify                    212
## 4 Unable to contact - Bad Contact Information    191
## 5 EP Confirmed                                   130
## 6 Contacted - Invalid Profile                     85
## 7 EP Sent                                         16
## 8 EP Received                                      2
```

All 7 non-null CallStatus values match the assignment definitions exactly, with no spelling errors, extra spaces, or unexpected categories. The 2,140 NA entries are confirmed as systemic across all months as established in Phase 1. This column is clean and ready for recoding in Phase 3 where NAs will become "No Outcome" and the 7 values will be collapsed into 4 groups: Closed, Good Quality (EP Sent/Received/Confirmed), Bad Quality (Unable to Contact, Invalid Profile, Doesn't Qualify), and No Outcome.

3 Flag the overlapping DebtLevel ranges

```
## # A tibble: 2 x 2
##   DebtLevel      n
##   <chr>      <int>
```

```
## 1 7500-10000   442
## 2 7500-15000   271
```

The overlap audit confirmed two categories sharing the same lower bound of $7,500: "7500-10000" (442 leads, 14.6%) and "7500-15000" (271 leads, 9.0%), totalling 713 leads or 23.6% of the dataset. This appears to be a form versioning issue where an earlier version of the ad collected a broader range ("7500-15000") and a later version split it into a narrower band ("7500-10000"). Since the raw data cannot confirm which is correct for each individual lead, both categories will be retained as-is in Phase 3 and collapsed into a single "Under $15,000" band in Phase 4 Feature Engineering. This prevents the segment from being artificially split during quality rate comparisons.

3.1 Data Quality Summary

Table 8: Phase 2: Data Quality Issues and Planned Fixes

Col	Issue	Action
Partner	Capitalisation inconsistency: 'google' vs 'Google' treated as separate partners	case_when() to manually standardise all 6 values
AddressScore	String 'NULL' stored instead of NA; column type is character not numeric	na_if('NULL') then as.numeric()
PhoneScore	String 'NULL' stored instead of NA; column type is character not numeric	na_if('NULL') then as.numeric()
IP Address	100% empty (3,021 NAs); logical type with no usable values	select(-'IP Address') - drop entirely
DebtLevel	Unordered character string; alphabetic sort is incorrect for range data	factor() with explicitly defined ordered levels
DebtLevel	Overlapping lower bound: '7500-10000' (442) and '7500-15000' (271) share same floor	Retain both in Phase 3; collapse to 'Under 15000' band in Phase 4
CallStatus	70.8% missing (2,140 rows); systemic across all 6 months	replace_na('No Outcome'); recode into 4 quality groups in Phase 4
Email	133 duplicate email addresses (4.4% of leads)	Flag duplicates; exclude from quality rate calculations in Phase 5
VendorLeadID	8 missing values	Flag only - retain rows

Phase 2 identified 7 data quality issues across 6 columns. No issues were found in State, CallStatus values, LeadCreated, or LandingPageURL. The most material issue for analysis is the Partner capitalisation inconsistency, which would have split Google's 53.6% share across two separate groups and significantly distorted partner-level quality comparisons. The IP Address column will be dropped entirely. All remaining issues have clearly defined fixes ready to execute in Phase 3.

4 Phase 3: Data Cleaning

4.1 Drop IP Address Column

```
## Columns after drop: 23
```

The IP Address column was removed, reducing the dataset from 24 to 23 columns. No analytical value is lost as the column was entirely empty across all 3,021 rows.

4.2 Standardise Partner Values

```
## # A tibble: 5 x 2
##   Partner           n
##   <chr>         <int>
## 1 Google         1618
## 2 Yahoo           958
## 3 Call Center     271
## 4 AdKnowledge     171
## 5 Advertise.com     3
```

Partner values are now clean across 5 categories. Google is the dominant traffic source at 1,618 leads (53.6%), followed by Yahoo at 958 (31.7%), Call Center at 271 (9.0%), AdKnowledge at 171 (5.7%), and Advertise.com at 3 (0.1%). The previous false split between "google" and "Google" is resolved.

4.3 Fix AddressScore and PhoneScore

```
## AddressScore NAs after fix: 1850

## PhoneScore NAs after fix: 1628

## AddressScore class: numeric

## PhoneScore class: numeric
```

Both columns are now correctly typed as numeric. AddressScore retains 1,850 NA values and PhoneScore retains 1,628, consistent with Phase 2 findings. No data was lost and no unexpected values were introduced by the conversion.

4.4 Recode CallStatus

```
## # A tibble: 3 x 2
##   QualityGroup     n
##   <chr>        <int>
## 1 No Outcome    2140
## 2 Bad            488
## 3 Good           393
```

All 3,021 leads are now assigned a QualityGroup with zero unclassified rows. No Outcome accounts for 2,140 leads (70.8%), Bad quality for 488 (16.2%), and Good quality for 393 (13.0%). The broad quality rate of 13.0% includes all EP pipeline leads alongside closed leads and is not directly comparable to the 9.6% contract target, which is defined on closed leads only. Section 4C uses the closed rate as the business performance metric.

4.5 Order DebtLevel Factor

```
## DebtLevel class: ordered factor

##  [1] "7500-10000"       "7500-15000"       "10001-15000"      "15001-20000"
##  [5] "20001-30000"      "30001-50000"      "50001-70000"      "70001-90000"
##  [9] "90000-100000"     "More_than_100000"
```

DebtLevel is now an ordered factor with all 10 levels correctly sequenced from lowest to highest debt band. Sorting and chart axis ordering in subsequent phases will reflect true debt magnitude rather than alphabetical order.

4.6 Flag Duplicate Emails

```
## Leads flagged as duplicate email: 164
```

```
## Unique leads (non-duplicate): 2857
```

164 leads share an email address with at least one other record, representing 5.4% of the dataset. These are flagged with is_duplicate_email = TRUE and retained in the dataset. The 2,857 non-duplicate leads will serve as the clean base for quality rate calculations in Phase 5.

Note: 133 is the count of distinct email addresses that appear more than once. 164 is the count of all rows flagged as non-first occurrences, since a single email appearing three times contributes 1 to the first count and 2 to the second. The 2,857 unique-lead base retains one row per email address and is the correct denominator for all quality calculations.

4.7 Verification

```
## Final row count: 3021
```

```
## Final column count: 25
```

```
## Remaining NAs in CallStatus: 0
```

```
## Remaining string NULLs in AddressScore: 0
```

```
## Partner values remaining: Google, AdKnowledge, Yahoo, Advertise.com, Call Center
```

```
## QualityGroup values: No Outcome, Bad, Good
```

All 8 issues identified in Phase 2 have been resolved. Row count holds at 3,021, confirming no records were dropped during cleaning. The column count of 25 reflects the original 24 columns, minus IP Address, plus the two newly engineered columns QualityGroup and is_duplicate_email. CallStatus has zero remaining NAs, AddressScore has zero remaining string NULLs, and all Partner and QualityGroup values are clean and consistent. The dataset is ready for Phase 4 Feature Engineering.

5 Phase 4: Feature Engineering

5.1 Time Features

```
## New time columns created: lead_month, lead_month_label, lead_week, lead_dow, lead_hour
```

```
## # A tibble: 6 x 2
##   lead_month_label     n
##   <chr>            <int>
## 1 2009-04            518
## 2 2009-05            580
## 3 2009-06            609
## 4 2009-07            532
## 5 2009-08            438
## 6 2009-09            344
```

Five time-based features were engineered from LeadCreated: lead_month (floor date to month), lead_month_label (formatted as YYYY-MM for readability), lead_week (floor date to week), lead_dow (day of week), and lead_hour (hour of day). Monthly distribution shows fairly consistent volume across the 6-month period, ranging from 344 leads in September to 609 in June, with no obvious seasonality.

5.2 Simplify Widget Name

```
## # A tibble: 9 x 2
##   widget_design        n
##   <chr>            <int>
## 1 CreditSolutions   1206
## 2 Standard           620
## 3 White              436
## 4 YellowArrow Blue   232
## 5 BlueMeter          179
## 6 YellowArrow Dark   135
## 7 Head2               89
## 8 Head3               75
## 9 YellowArrow         49
```

```
## # A tibble: 2 x 2
##   widget_pages     n
##   <chr>        <int>
## 1 1-Page Form   2859
## 2 2-Page Form    162
```

Widget designs were collapsed from 14 unique widget name strings into 9 design categories plus a form structure flag (1-page vs 2-page). CreditSolutions is the dominant design at 1,206 leads (39.9%), followed by Standard at 620 (20.5%) and White at 436 (14.4%). The YellowArrow variants combined account for 416 leads (13.8%). One-page forms dominate at 2,859 leads (94.6%), with only 162 two-page forms (5.4%).

5.3 Collapse DebtLevel Bands

```
## # A tibble: 8 x 2
##   DebtBand      n
##   <ord>     <int>
## 1 Under 15K  1004
## 2 15K-20K     408
## 3 20K-30K     456
## 4 30K-50K     496
## 5 50K-70K     245
## 6 70K-90K     131
```

```
## 7 90K-100K     90
## 8 Over 100K   191
```

The original 10 DebtLevel categories were collapsed into 8 ordered debt bands using intuitive $5K increments. The "Under 15K" band absorbed the overlapping 7500-10000, 7500-15000, and 10001-15000 ranges, totalling 1,004 leads (33.2%). This prevents artificial segment splits during quality rate comparisons. All other bands maintain clean, non-overlapping ranges.

5.4 Branded vs Generic Flag

```
## # A tibble: 2 x 2
##   ad_type     n
##   <chr>   <int>
## 1 Branded  1206
## 2 Generic  1815
```

Ad type split shows Branded (CreditSolutions) at 1,206 leads (39.9%) versus Generic (Debt Settlement Master) at 1,815 leads (60.1%), representing a fairly balanced split across the dataset. This binary split will support straightforward quality rate comparison between advertiser branding strategies.

5.5 Online vs Call Center Flag

```
## # A tibble: 2 x 2
##   lead_source     n
##   <chr>       <int>
## 1 Call Center   271
## 2 Online       2750
```

"Lead source produces two values: Call Center (271 leads, 9.0%) and Online (2,750 leads, 91.0%). The 271 Call Center leads were generated by phone, with form data entered by call center staff on behalf of the consumer."

5.6 Lead Verification Score

```
## verify_score NAs: 1628
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   1.500   3.500   4.000   4.058   5.000   5.000    1628
```

A composite verification score was engineered by averaging AddressScore and PhoneScore where available. The score ranges from 1.5 to 5.0 with a median of 4.0 and mean of 4.058, indicating generally high address and phone verification across the leads that have both scores. However, 1,628 leads (53.9%) have a missing verify_score because they lack PhoneScore data, a limitation of the data collection period.

5.7 Save Clean Dataset

```
## Clean dataset saved to data/clean_data_final.xlsx
```

```
## Rows: 3021
```

```
## Columns: 36
```

The clean engineered dataset was saved to Excel format (clean_data_final.xlsx) with 3,021 rows and 36 columns. The dataset includes all original raw columns except IP Address, plus 11 newly engineered features: QualityGroup, is_duplicate_email, lead_month, lead_month_label, lead_week, lead_dow, lead_hour, widget_pages, widget_design, DebtBand, ad_type, lead_source, and verify_score.

5.8 Verification

```
## Total columns now: 36

## New columns added in Phase 4: lead_month, lead_month_label, lead_week, lead_dow, lead_hour, widget_pa

##
## QualityGroup distribution:

## # A tibble: 3 x 2
##   QualityGroup     n
##   <chr>        <int>
## 1 Bad            488
## 2 Good           393
## 3 No Outcome    2140

##
## Ad Type distribution:

## # A tibble: 2 x 2
##   ad_type     n
##   <chr>   <int>
## 1 Branded  1206
## 2 Generic  1815

##
## Lead Source distribution:

## # A tibble: 2 x 2
##   lead_source     n
##   <chr>       <int>
## 1 Call Center   271
## 2 Online       2750
```

All 11 engineered features are now present in the dataset. QualityGroup shows the expected distribution: No Outcome (2,140, 70.8%), Bad (488, 16.2%), and Good (393, 13.0%). Ad Type is split 39.9% Branded vs 60.1% Generic. The dataset is clean, feature-rich, and ready for Phase 5 Exploratory Data Analysis.

6 Phase 5: Exploratory Data Analysis

6.1 Analysis Setup

```
## Base dataset (non-duplicate, all outcomes): 2857 leads
```

```
## Good leads: 372
```

```
## Bad leads: 443
```

```
## No Outcome leads: 2042
```

The base dataset for quality rate analysis contains 2,857 non-duplicate leads after excluding 164 flagged duplicate email records. Of these, 372 are Good quality (13.0%), 443 are Bad quality (15.5%), and 2,042 have No Outcome (71.5%). All quality rates in this phase are calculated as Good / Total including No Outcome leads, which is the correct denominator for business performance measurement.

6.2 Overall Baseline Quality Rate

Table 9: Overall Baseline Quality Rate

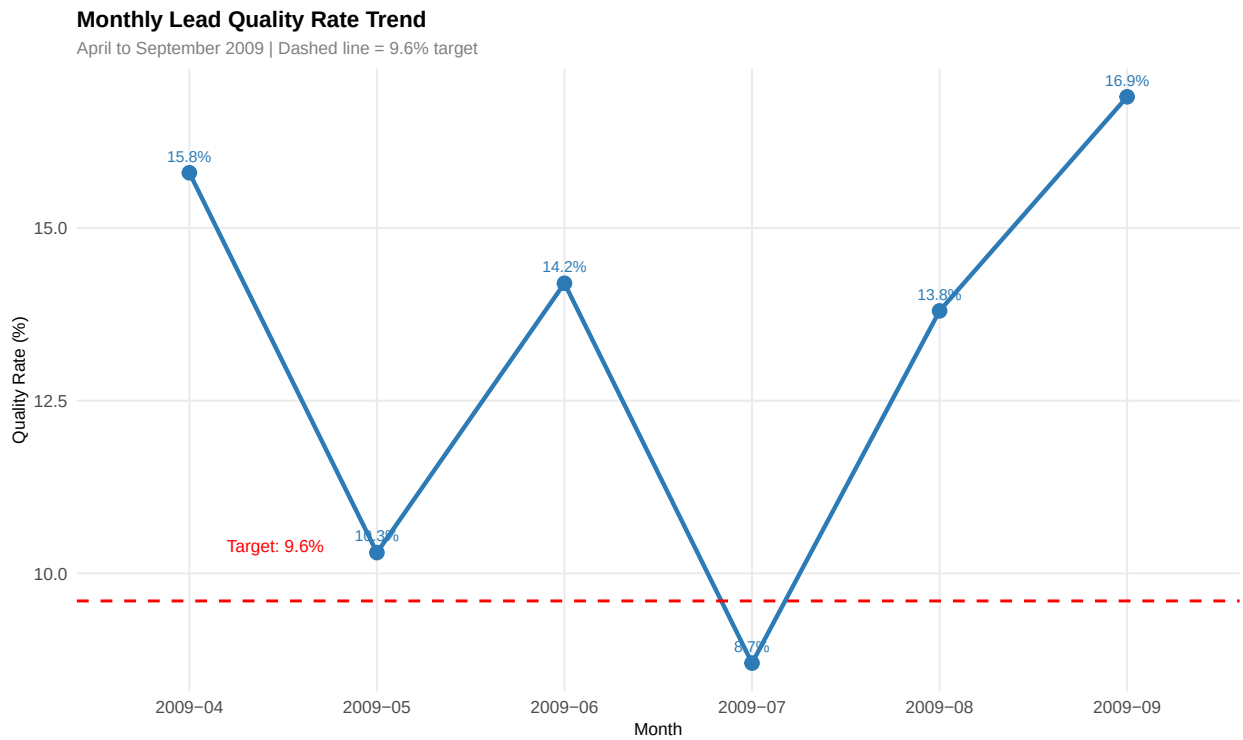
Total_Leads	Good_Leads	Bad_Leads	Quality_Rate	Target	Gap
2857	372	443	13	9.6	-3.4

The overall quality rate stands at 13.0% against a target of 9.6%, indicating the campaign is currently performing 3.4 percentage points above target. However, this headline number masks significant variation across segments. The No Outcome rate of 71.5% means that nearly 3 in every 4 leads are never followed up on, which represents the single largest opportunity for improvement across the entire dataset. This 13.0% figure is the broad quality rate (Good = Closed + EP steps). The Phase 8 business impact analysis uses Closed-only leads as the contract metric, which produces an 8.09% baseline.

6.3 Quality Rate by Month

Table 10: Quality Rate by Month

lead_month_label	Total	Good	Bad	No_Outcome	Rate
2009-04	487	77	53	357	15.8
2009-05	554	57	52	445	10.3
2009-06	570	81	127	362	14.2
2009-07	494	43	95	356	8.7
2009-08	421	58	61	302	13.8
2009-09	331	56	55	220	16.9

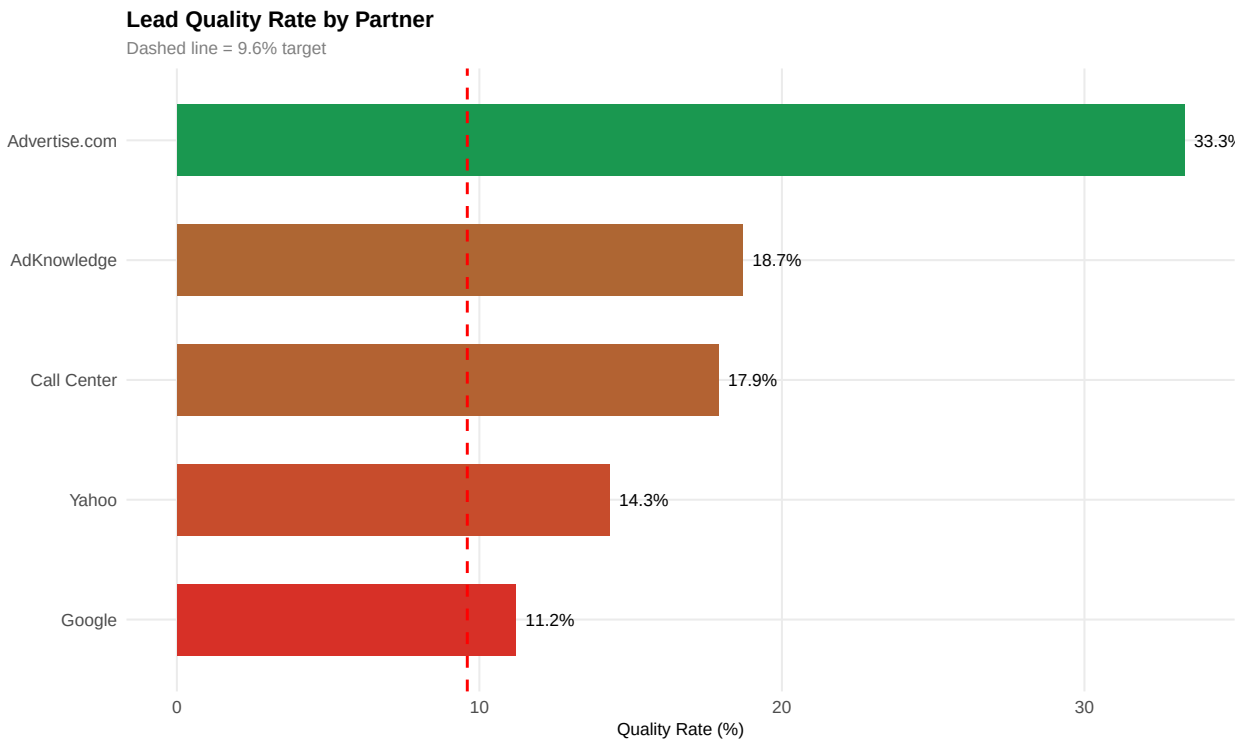


Quality rate shows an uneven pattern across the 6-month period, ranging from a low of 8.7% in July to a high of 16.9% in September. July is the only month that falls below the 9.6% target, while April (15.8%), June (14.2%), August (13.8%), and September (16.9%) all exceed it. There is no consistent upward or downward trend, which suggests quality is influenced by segment mix changes month to month rather than a structural improvement or decline. This will be tested formally in Phase 7 with Spearman correlation.

6.4 Quality Rate by Partner

Table 11: Quality Rate by Partner

Partner	Total	Good	Bad	No_Outcome	Rate
Advertise.com	3	1	0	2	33.3
AdKnowledge	171	32	23	116	18.7
Call Center	134	24	27	83	17.9
Yahoo	949	136	146	667	14.3
Google	1600	179	247	1174	11.2

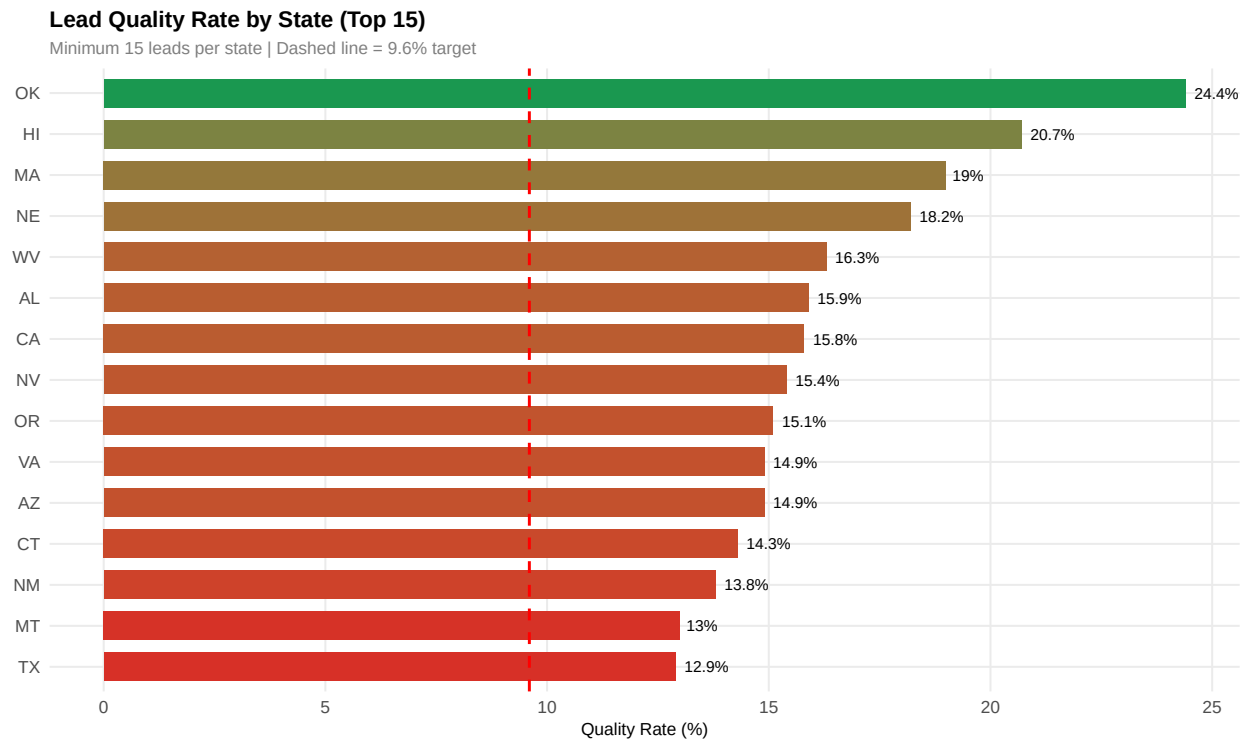


AdKnowledge delivers the highest meaningful quality rate at 18.7% from 171 leads, followed by Call Center at 17.9% from 134 leads and Yahoo at 14.3% from 949 leads. Google, despite being the dominant traffic source at 1,600 leads (56.0% of the base), produces the lowest quality rate among meaningful partners at 11.2%. Advertise.com's 33.3% rate is based on only 3 leads and is not statistically actionable. The gap between AdKnowledge (18.7%) and Google (11.2%) is 7.5 percentage points, which at Google's volume represents a significant number of missed good leads.

6.5 Quality Rate by State

Table 12: Quality Rate by State (Top 15, min 15 leads)

State	Total	Good	Bad	No_Outcome	Rate
OK	45	11	8	26	24.4
HI	29	6	5	18	20.7
MA	63	12	12	39	19.0
NE	22	4	3	15	18.2
WV	49	8	7	34	16.3
AL	88	14	12	62	15.9
CA	438	69	65	304	15.8
NV	52	8	9	35	15.4
OR	53	8	11	34	15.1
AZ	87	13	19	55	14.9
VA	141	21	16	104	14.9
CT	21	3	3	15	14.3
NM	29	4	7	18	13.8
MT	23	3	5	15	13.0
TX	310	40	39	231	12.9

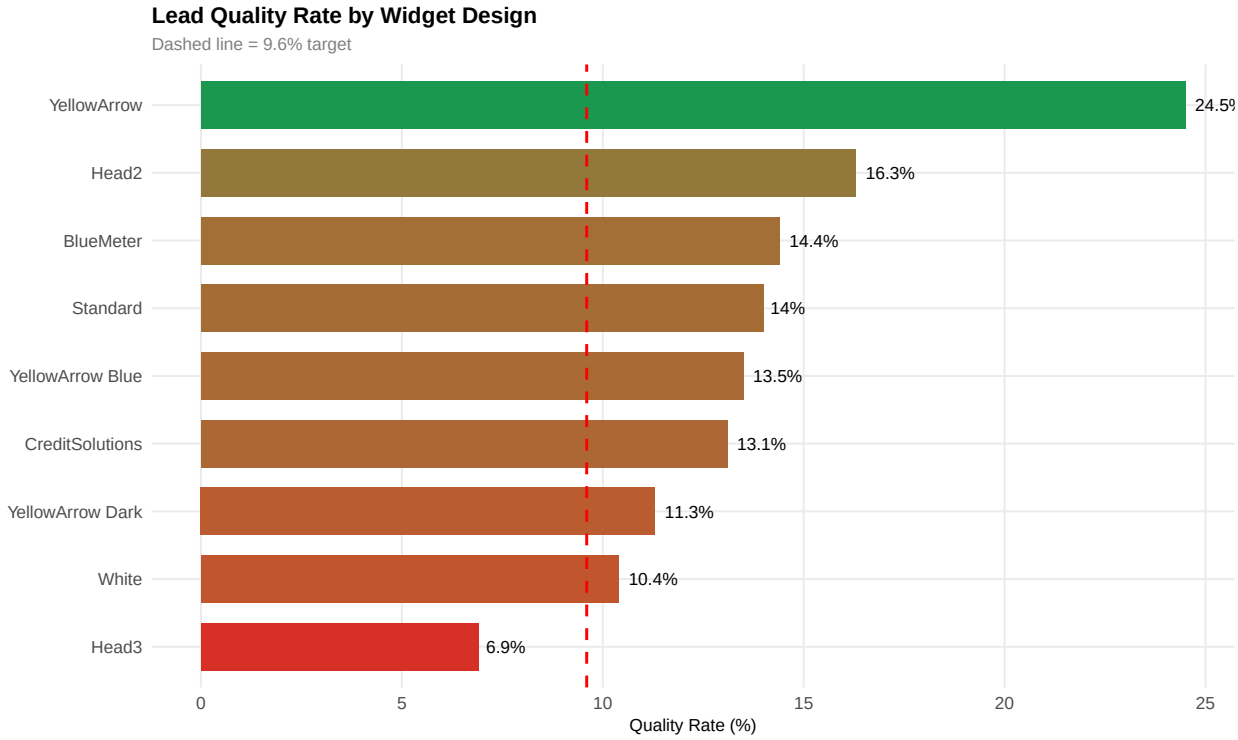


Oklahoma leads all states with a 24.4% quality rate from 45 leads, followed by Hawaii at 20.7% and Massachusetts at 19.0%. California and Texas, the two highest volume states at 438 and 310 leads respectively, both perform above target at 15.8% and 12.9%. The top 5 states by quality rate all exceed the 9.6% target by at least 3 percentage points, suggesting geographic targeting adjustments could meaningfully lift overall performance. Note: Only states with a minimum of 15 leads are included to ensure statistical reliability.

6.6 Quality Rate by Widget Design

Table 13: Quality Rate by Widget Design

widget_design	Total	Good	Bad	No_Outcome	Rate
YellowArrow	49	12	9	28	24.5
Head2	86	14	8	64	16.3
BlueMeter	174	25	13	136	14.4
Standard	486	68	79	339	14.0
YellowArrow Blue	229	31	48	150	13.5
CreditSolutions	1196	157	188	851	13.1
YellowArrow Dark	133	15	24	94	11.3
White	432	45	67	320	10.4
Head3	72	5	7	60	6.9

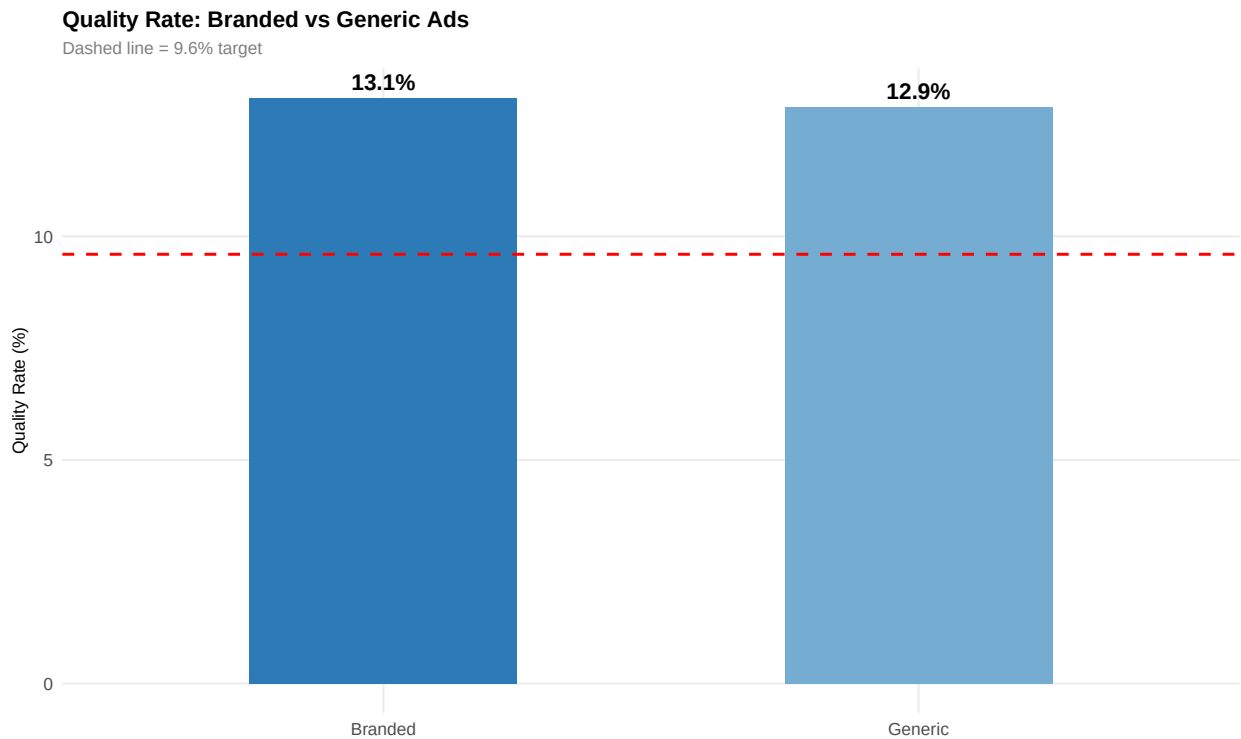


YellowArrow achieves the highest quality rate at 24.5% but from only 49 leads, limiting its statistical weight. The two most analytically reliable high performers are Head2 at 16.3% (86 leads) and BlueMeter at 14.4% (174 leads). The dominant widget CreditSolutions runs at 13.1% across 1,196 leads, performing slightly above target. Head3 is the clear underperformer at 6.9% from 72 leads, the only widget below the 9.6% target with sufficient volume to be meaningful. Shifting budget away from Head3 toward BlueMeter or Head2 is a directly actionable lever.

6.7 Quality Rate by Ad Type

Table 14: Quality Rate by Ad Type: Branded vs Generic

ad_type	Total	Good	Bad	No_Outcome	Rate
Branded	1196	157	188	851	13.1
Generic	1661	215	255	1191	12.9

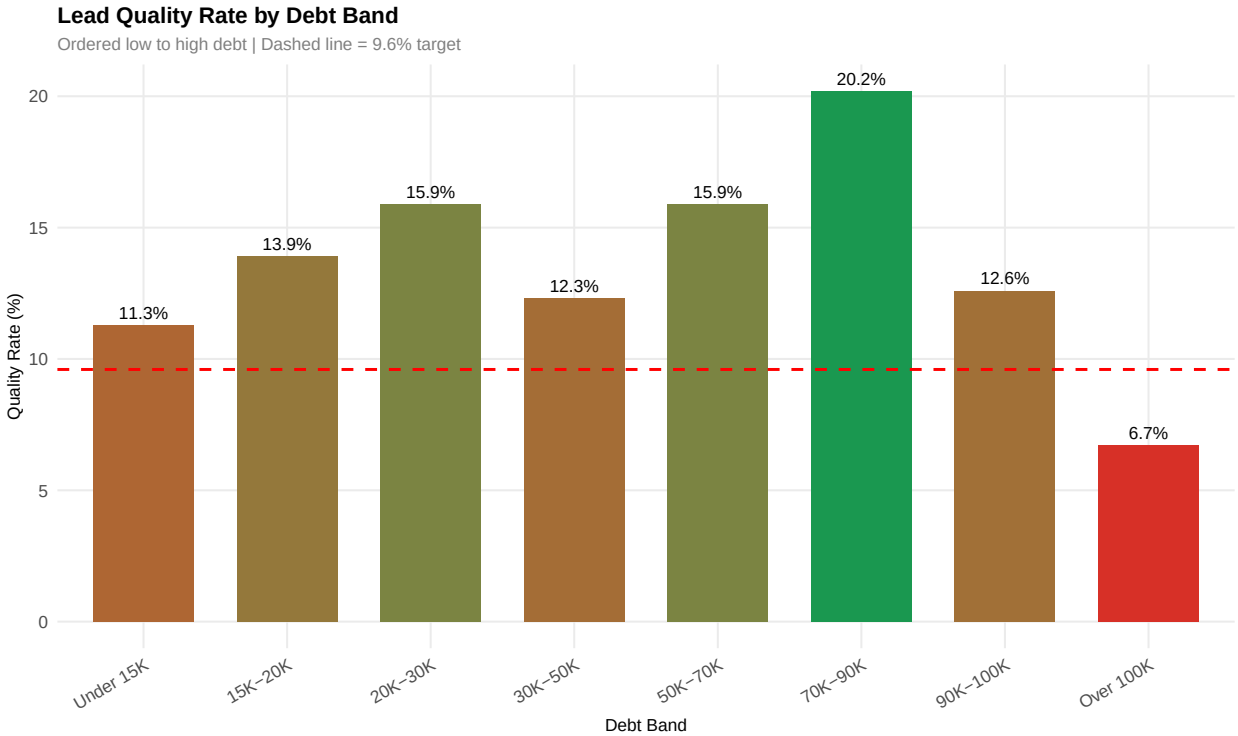


Branded and Generic ads perform nearly identically at 13.1% and 12.9% respectively, a difference of only 0.2 percentage points. Both exceed the 9.6% target. This finding indicates that ad type alone is not a meaningful differentiator of lead quality in this dataset. The more important question for Phase 6 is whether Branded or Generic performs differently within specific partners or debt bands, where interaction effects may reveal hidden variation.

6.8 Quality Rate by Debt Band

Table 15: Quality Rate by Debt Band

DebtBand	Total	Good	Bad	No_Outcome	Rate
Under 15K	940	106	165	669	11.3
15K-20K	380	53	44	283	13.9
20K-30K	441	70	66	305	15.9
30K-50K	472	58	79	335	12.3
50K-70K	233	37	29	167	15.9
70K-90K	124	25	15	84	20.2
90K-100K	87	11	12	64	12.6
Over 100K	180	12	33	135	6.7

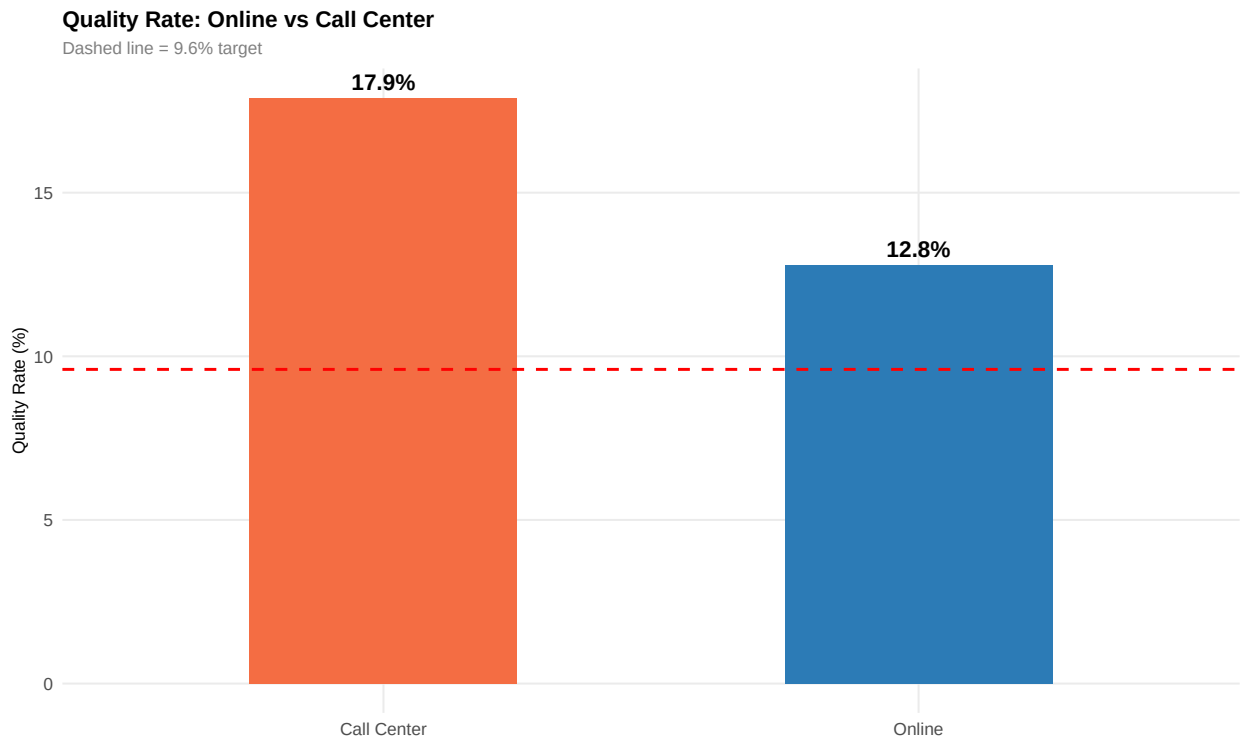


Quality rate follows a non-linear pattern across debt bands. The 70K-90K band delivers the highest rate at 20.2% from 124 leads, while Over 100K delivers the lowest at 6.7% from 180 leads, the only debt band below target with material volume. The mid-range bands of 20K-30K and 50K-70K both perform at 15.9%. The Under 15K band, which is the largest segment at 940 leads, runs at 11.3%, slightly above target. The Over 100K band's underperformance warrants investigation in Phase 6 to determine whether it is concentrated in a specific partner or widget.

6.9 Quality Rate by Lead Source

Table 16: Quality Rate by Lead Source

lead_source	Total	Good	Bad	No_Outcome	Rate
Call Center	134	24	27	83	17.9
Online	2723	348	416	1959	12.8



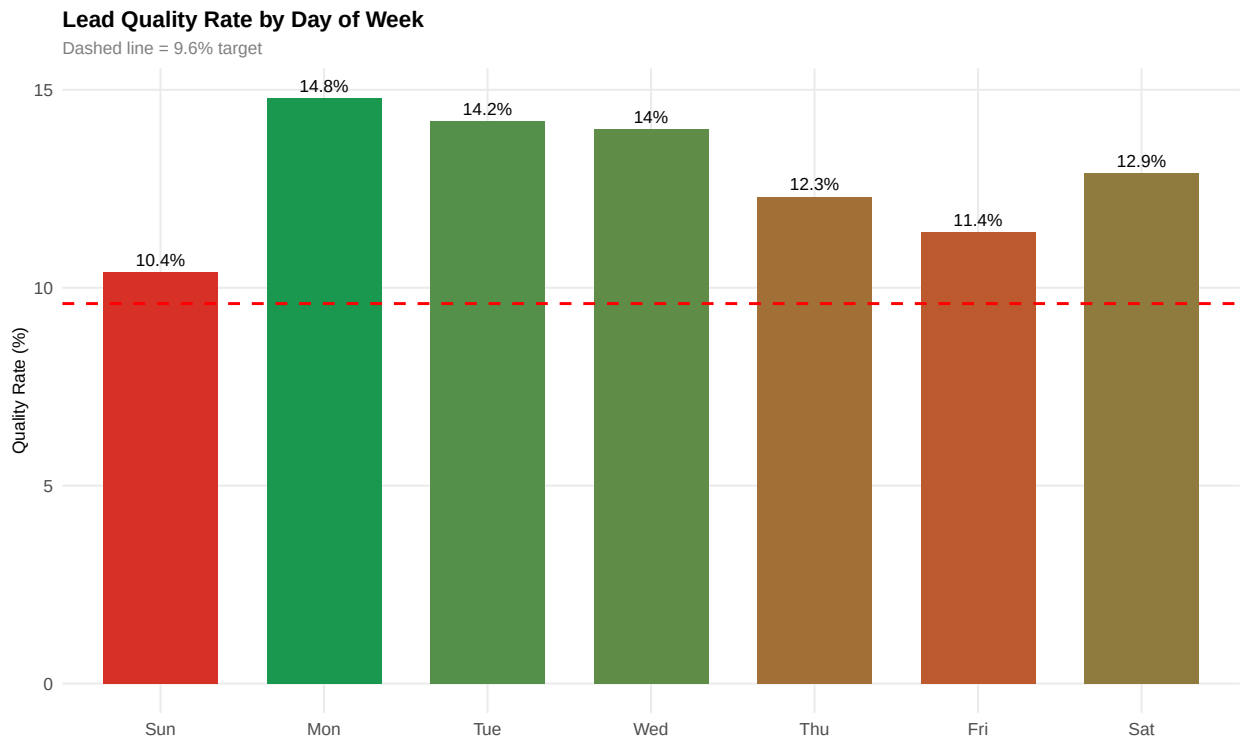
Call Center leads outperform Online leads at 17.9% versus 12.8%, a gap of 5.1 percentage points. This is expected: Call Center staff pre-screen leads before entry, whereas online leads are self-submitted with no upfront qualification. Despite Call Center's higher quality rate, it accounts for only 134 leads (4.7% of the base), limiting its total impact on the overall rate. Online leads at 2,723 records drive the vast majority of volume and represent the primary lever for improving overall quality.

Note: Of the original 271 Call Center leads, 137 (50.6%) were flagged as duplicate emails in Phase 3, a rate three times higher than the Online channel (0.9%). This is consistent with call center agents entering the same consumer multiple times across repeated calls.

6.10 Quality Rate by Day of Week

Table 17: Quality Rate by Day of Week

lead_dow	Total	Good	Bad	No_Outcome	Rate
Sun	346	36	40	270	10.4
Mon	466	69	82	315	14.8
Tue	486	69	78	339	14.2
Wed	437	61	81	295	14.0
Thu	381	47	56	278	12.3
Fri	385	44	65	276	11.4
Sat	356	46	41	269	12.9



Monday and Tuesday are the strongest performing days at 14.8% and 14.2% respectively, while Sunday and Friday are the weakest at 10.4% and 11.4%. All days exceed the 9.6% target. The 4.4 percentage point gap between Monday and Sunday is the widest day-of-week variation observed. This pattern is consistent with typical B2C lead generation, where early-week intent is higher. The variation is relatively modest compared to partner or debt band differences, making day of week a secondary optimisation lever.

6.11 EDA Summary

Table 18: Phase 5: Full EDA Summary - All Segments Ranked by Quality Rate

Dimension	Category	Total	Good	Bad	Rate
Partner	Advertise.com	3	1	0	33.3
Widget Design	YellowArrow	49	12	9	24.5
Debt Band	70K-90K	124	25	15	20.2
Partner	AdKnowledge	171	32	23	18.7
Partner	Call Center	134	24	27	17.9
Lead Source	Call Center	134	24	27	17.9
Widget Design	Head2	86	14	8	16.3
Debt Band	20K-30K	441	70	66	15.9
Debt Band	50K-70K	233	37	29	15.9
Widget Design	BlueMeter	174	25	13	14.4
Partner	Yahoo	949	136	146	14.3
Widget Design	Standard	486	68	79	14.0
Debt Band	15K-20K	380	53	44	13.9
Widget Design	YellowArrow Blue	229	31	48	13.5
Widget Design	CreditSolutions	1196	157	188	13.1
Ad Type	Branded	1196	157	188	13.1

Dimension	Category	Total	Good	Bad	Rate
Ad Type	Generic	1661	215	255	12.9
Lead Source	Online	2723	348	416	12.8
Debt Band	90K-100K	87	11	12	12.6
Debt Band	30K-50K	472	58	79	12.3
Widget Design	YellowArrow Dark	133	15	24	11.3
Debt Band	Under 15K	940	106	165	11.3
Partner	Google	1600	179	247	11.2
Widget Design	White	432	45	67	10.4
Widget Design	Head3	72	5	7	6.9
Debt Band	Over 100K	180	12	33	6.7

Across all segments, the strongest quality rate signals come from three dimensions: Partner (AdKnowledge at 18.7% vs Google at 11.2%), Debt Band (70K-90K at 20.2% vs Over 100K at 6.7%), and Widget Design (Head3 at 6.9% as a clear underperformer). Ad Type shows no meaningful differentiation. These three dimensions will be the primary focus of Phase 6 multi-variable analysis, where combinations of top-performing partner, debt band, and widget design segments will be examined for compounding quality rate effects. Note: Call Center appears under both Partner and Lead Source dimensions as it represents the same 134 leads in both segmentations.

7 Phase 6: Multi-Variable Analysis

7.1 Setup

```
## Phase 6 setup complete. calc_quality_2d() ready.
```

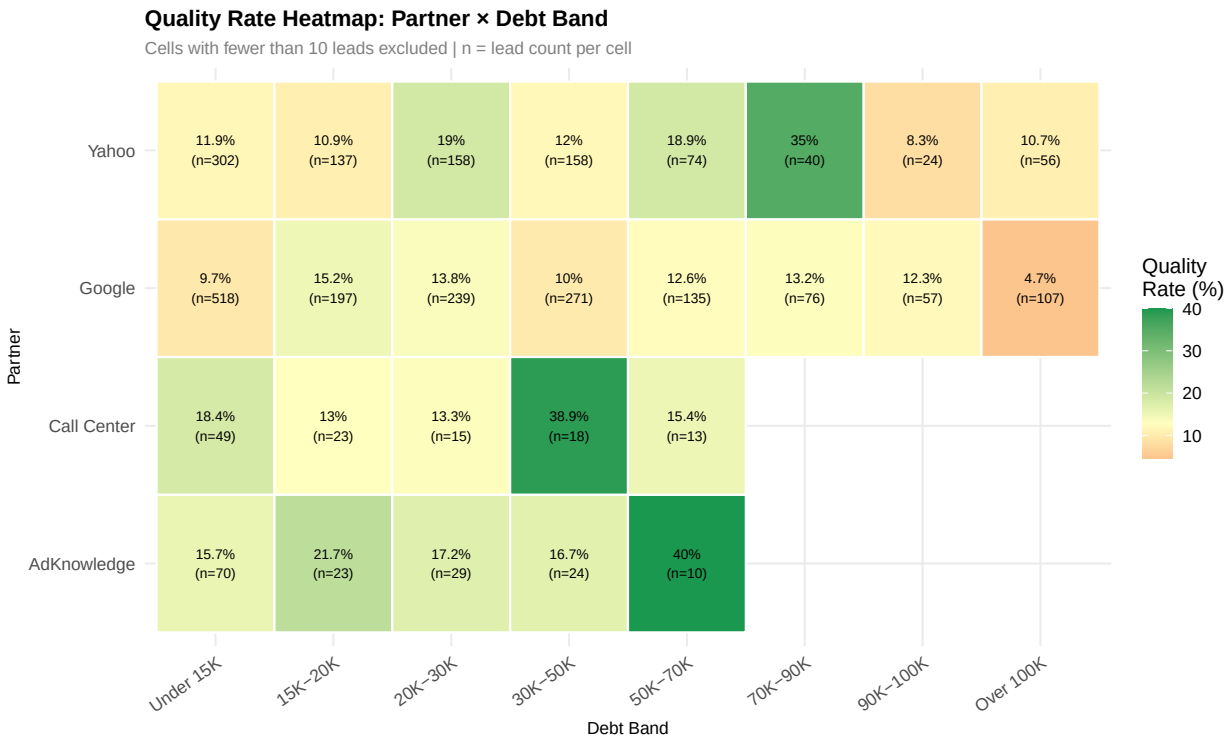
The calc_quality_2d() function extends Phase 5 single-dimension analysis into two-way combinations, grouping leads by two segments simultaneously and applying a minimum threshold of 10 leads per cell to exclude statistically unreliable intersections. Advertise.com is excluded from all partner combinations due to its 3-lead total, which produces meaningless rates at any intersection.

7.2 Partner × Debt Band

Table 19: Quality Rate: Partner × Debt Band (min 10 leads)

Partner	DebtBand	Total	Good	Bad	No_Outcome	Rate
AdKnowledge	Under 15K	70	11	11	48	15.7
AdKnowledge	15K-20K	23	5	2	16	21.7
AdKnowledge	20K-30K	29	5	7	17	17.2
AdKnowledge	30K-50K	24	4	1	19	16.7
AdKnowledge	50K-70K	10	4	0	6	40.0
Call Center	Under 15K	49	9	11	29	18.4
Call Center	15K-20K	23	3	5	15	13.0
Call Center	20K-30K	15	2	1	12	13.3
Call Center	30K-50K	18	7	4	7	38.9
Call Center	50K-70K	13	2	0	11	15.4
Google	Under 15K	518	50	92	376	9.7
Google	15K-20K	197	30	22	145	15.2

Partner	DebtBand	Total	Good	Bad	No_Outcome	Rate
Google	20K-30K	239	33	38	168	13.8
Google	30K-50K	271	27	43	201	10.0
Google	50K-70K	135	17	16	102	12.6
Google	70K-90K	76	10	8	58	13.2
Google	90K-100K	57	7	8	42	12.3
Google	Over 100K	107	5	20	82	4.7
Yahoo	Under 15K	302	36	51	215	11.9
Yahoo	15K-20K	137	15	15	107	10.9
Yahoo	20K-30K	158	30	20	108	19.0
Yahoo	30K-50K	158	19	31	108	12.0
Yahoo	50K-70K	74	14	13	47	18.9
Yahoo	70K-90K	40	14	5	21	35.0
Yahoo	90K-100K	24	2	4	18	8.3
Yahoo	Over 100K	56	6	7	43	10.7

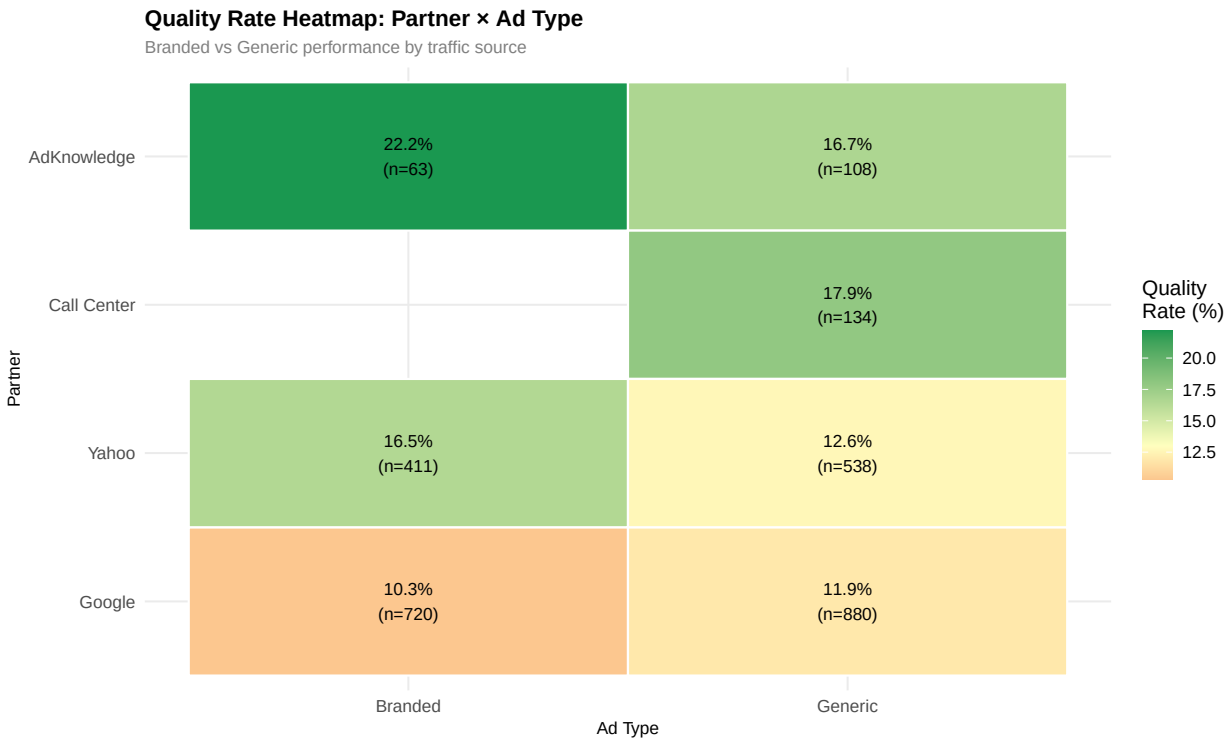


The most actionable finding in this entire analysis is the Google + Over 100K combination, which delivers only 4.7% quality rate from 107 leads, the lowest rate of any meaningful cell in the dataset and the only partner-debt combination that falls materially below the 9.6% target at volume. By contrast, Yahoo + 70K-90K produces 35.0% from 40 leads, and AdKnowledge + 50K-70K produces 40.0% from 10 leads, though both have limited sample size. For Google, the sweet spot is the 15K-20K band at 15.2% from 197 leads, making it the highest-volume, above-target cell within Google's portfolio. The practical recommendation: shift Google budget away from the Over 100K segment entirely.

7.3 Partner × Ad Type

Table 20: Quality Rate: Partner × Ad Type (min 10 leads)

Partner	ad_type	Total	Good	Bad	No_Outcome	Rate
AdKnowledge	Branded	63	14	14	35	22.2
AdKnowledge	Generic	108	18	9	81	16.7
Call Center	Generic	134	24	27	83	17.9
Google	Branded	720	74	112	534	10.3
Google	Generic	880	105	135	640	11.9
Yahoo	Branded	411	68	62	281	16.5
Yahoo	Generic	538	68	84	386	12.6



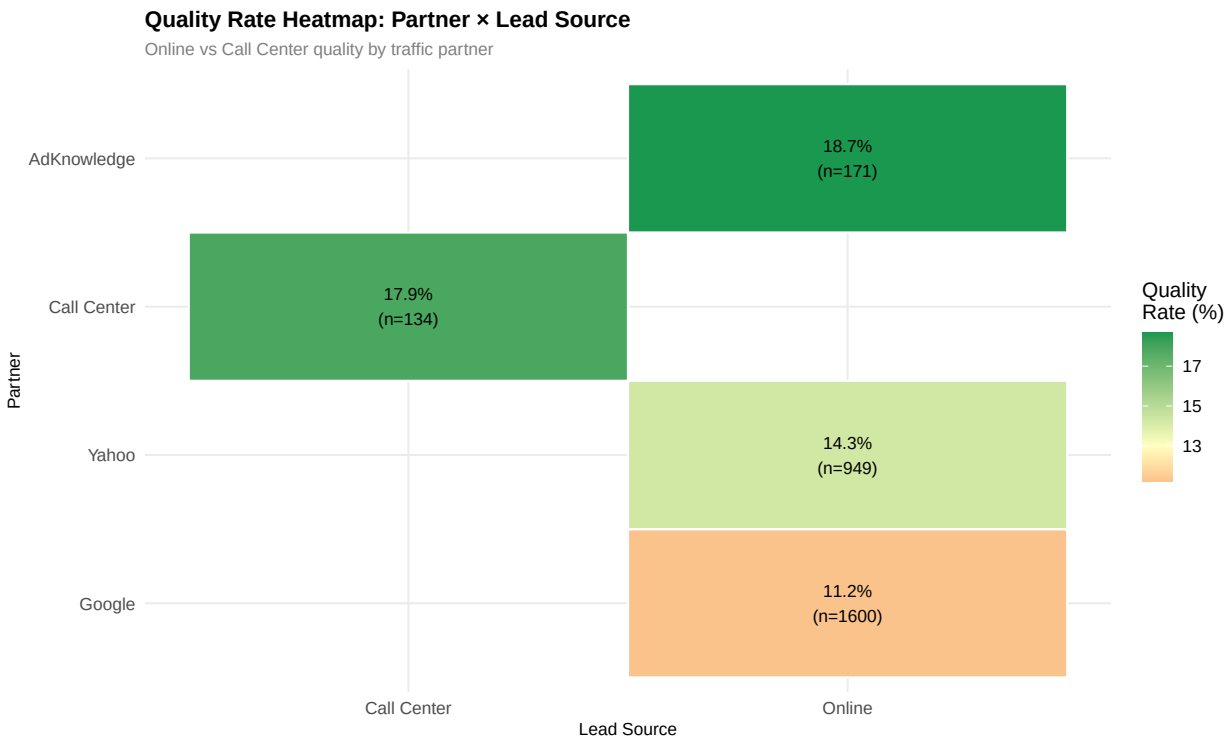
The Branded vs Generic distinction matters significantly within Yahoo but not within Google. Yahoo Branded delivers 16.5% from 411 leads versus Yahoo Generic at 12.6% from 538 leads, a 3.9 percentage point gap at high volume. AdKnowledge Branded leads at 22.2% from 63 leads is the highest rate in this matrix. Google Branded underperforms Generic at 10.3% versus 11.9%, a reversal of the expected pattern, suggesting the CreditSolutions branded widget does not resonate as strongly with Google's traffic as it does with Yahoo's audience. Call Center has no Branded leads, confirming all Call Center traffic goes through Generic campaigns only.

7.4 Partner × Lead Source

Table 21: Quality Rate: Partner × Lead Source (min 10 leads)

Partner	lead_source	Total	Good	Bad	No_Outcome	Rate
AdKnowledge	Online	171	32	23	116	18.7
Call Center	Call Center	134	24	27	83	17.9
Google	Online	1600	179	247	1174	11.2

Partner	lead_source	Total	Good	Bad	No_Outcome	Rate
Yahoo	Online	949	136	146	667	14.3



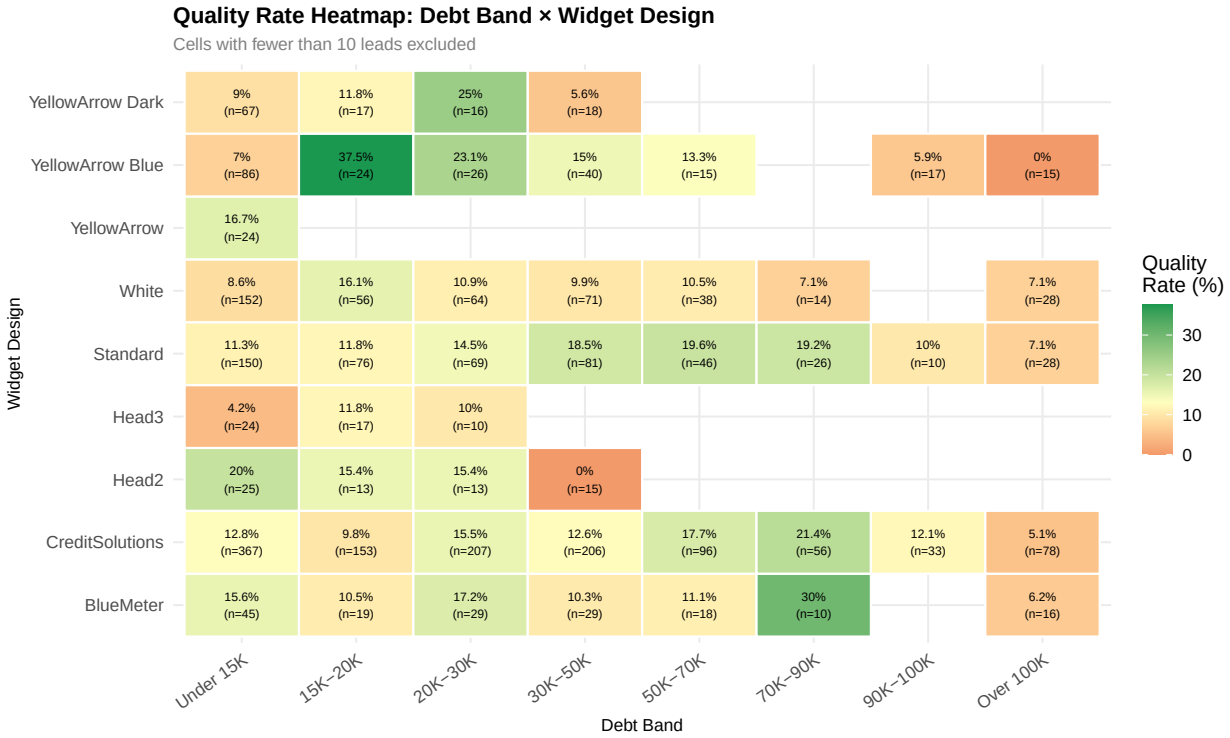
This matrix confirms structural data: AdKnowledge, Google, and Yahoo are exclusively Online partners, while Call Center operates as its own distinct channel with no overlap with other partners. There are no cross-partner, cross-source combinations to evaluate. The key analytical value here is confirming that Call Center's 17.9% quality rate is channel-specific and cannot be attributed to a particular advertising partner, meaning it is driven by the phone-based pre-screening process rather than traffic source quality.

7.5 Debt Band × Widget Design

Table 22: Quality Rate: Debt Band × Widget Design (min 10 leads)

DebtBand	widget_design	Total	Good	Bad	No_Outcome	Rate
Under 15K	BlueMeter	45	7	6	32	15.6
Under 15K	CreditSolutions	367	47	58	262	12.8
Under 15K	Head2	25	5	4	16	20.0
Under 15K	Head3	24	1	1	22	4.2
Under 15K	Standard	150	17	29	104	11.3
Under 15K	White	152	13	29	110	8.6
Under 15K	YellowArrow	24	4	2	18	16.7
Under 15K	YellowArrow Blue	86	6	23	57	7.0
Under 15K	YellowArrow Dark	67	6	13	48	9.0
15K-20K	BlueMeter	19	2	1	16	10.5
15K-20K	CreditSolutions	153	15	22	116	9.8
15K-20K	Head2	13	2	0	11	15.4

DebtBand	widget_design	Total	Good	Bad	No_Outcome	Rate
15K-20K	Head3	17	2	2	13	11.8
15K-20K	Standard	76	9	9	58	11.8
15K-20K	White	56	9	3	44	16.1
15K-20K	YellowArrow Blue	24	9	3	12	37.5
15K-20K	YellowArrow Dark	17	2	3	12	11.8
20K-30K	BlueMeter	29	5	0	24	17.2
20K-30K	CreditSolutions	207	32	31	144	15.5
20K-30K	Head2	13	2	3	8	15.4
20K-30K	Head3	10	1	0	9	10.0
20K-30K	Standard	69	10	14	45	14.5
20K-30K	White	64	7	10	47	10.9
20K-30K	YellowArrow Blue	26	6	4	16	23.1
20K-30K	YellowArrow Dark	16	4	2	10	25.0
30K-50K	BlueMeter	29	3	2	24	10.3
30K-50K	CreditSolutions	206	26	36	144	12.6
30K-50K	Head2	15	0	1	14	0.0
30K-50K	Standard	81	15	13	53	18.5
30K-50K	White	71	7	12	52	9.9
30K-50K	YellowArrow Blue	40	6	10	24	15.0
30K-50K	YellowArrow Dark	18	1	4	13	5.6
50K-70K	BlueMeter	18	2	1	15	11.1
50K-70K	CreditSolutions	96	17	15	64	17.7
50K-70K	Standard	46	9	2	35	19.6
50K-70K	White	38	4	5	29	10.5
50K-70K	YellowArrow Blue	15	2	3	10	13.3
70K-90K	BlueMeter	10	3	0	7	30.0
70K-90K	CreditSolutions	56	12	7	37	21.4
70K-90K	Standard	26	5	2	19	19.2
70K-90K	White	14	1	1	12	7.1
90K-100K	CreditSolutions	33	4	7	22	12.1
90K-100K	Standard	10	1	1	8	10.0
90K-100K	YellowArrow Blue	17	1	2	14	5.9
Over 100K	BlueMeter	16	1	2	13	6.2
Over 100K	CreditSolutions	78	4	12	62	5.1
Over 100K	Standard	28	2	9	17	7.1
Over 100K	White	28	2	6	20	7.1
Over 100K	YellowArrow Blue	15	0	2	13	0.0



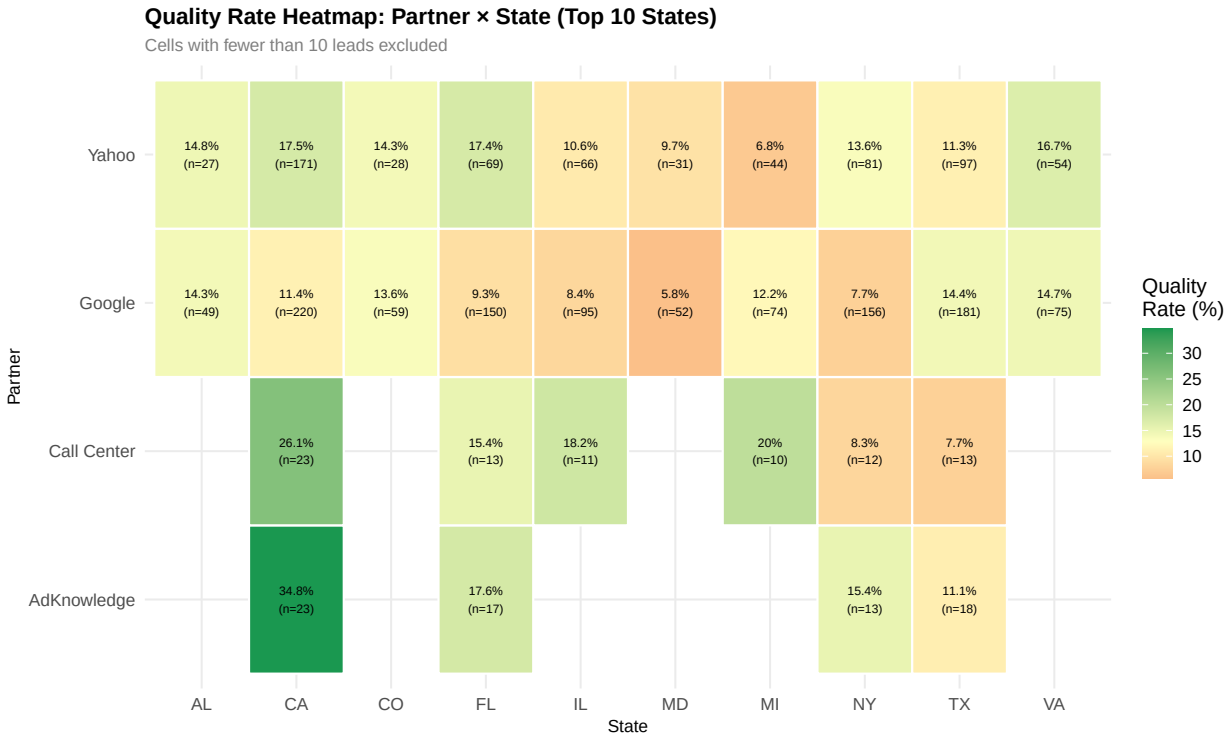
The single most striking cell is YellowArrow Blue + 15K-20K at 37.5% quality rate from 24 leads. While the volume is modest, this is directionally meaningful. The structural finding is that CreditSolutions widget, which accounts for the majority of impressions across all debt bands, consistently underperforms in the Over 100K segment at 5.1% from 78 leads, confirming this combination as the primary drag on overall quality. The Standard widget performs well in mid-to-high debt bands: 18.5% in 30K-50K and 19.6% in 50K-70K, suggesting it resonates better with higher-intent, higher-debt prospects. Head3 should be deprioritised in Under 15K (4.2%) and YellowArrow Blue in Over 100K (0.0% from 15 leads).

7.6 Partner × State (Top 10 States)

Table 23: Quality Rate: Partner × State - Top 10 States (min 10 leads)

Partner	State	Total	Good	Bad	No_Outcome	Rate
AdKnowledge	CA	23	8	3	12	34.8
AdKnowledge	FL	17	3	1	13	17.6
AdKnowledge	NY	13	2	3	8	15.4
AdKnowledge	TX	18	2	1	15	11.1
Call Center	CA	23	6	4	13	26.1
Call Center	FL	13	2	2	9	15.4
Call Center	IL	11	2	3	6	18.2
Call Center	MI	10	2	2	6	20.0
Call Center	NY	12	1	0	11	8.3
Call Center	TX	13	1	1	11	7.7
Google	AL	49	7	6	36	14.3
Google	CA	220	25	30	165	11.4
Google	CO	59	8	9	42	13.6
Google	FL	150	14	23	113	9.3

Partner	State	Total	Good	Bad	No_Outcome	Rate
Google	IL	95	8	7	80	8.4
Google	MD	52	3	12	37	5.8
Google	MI	74	9	11	54	12.2
Google	NY	156	12	32	112	7.7
Google	TX	181	26	25	130	14.4
Google	VA	75	11	9	55	14.7
Yahoo	AL	27	4	2	21	14.8
Yahoo	CA	171	30	28	113	17.5
Yahoo	CO	28	4	4	20	14.3
Yahoo	FL	69	12	12	45	17.4
Yahoo	IL	66	7	13	46	10.6
Yahoo	MD	31	3	3	25	9.7
Yahoo	MI	44	3	7	34	6.8
Yahoo	NY	81	11	9	61	13.6
Yahoo	TX	97	11	12	74	11.3
Yahoo	VA	54	9	5	40	16.7



AdKnowledge + California is the standout combination at 34.8% quality rate from 23 leads, more than double the overall baseline. Yahoo + California at 17.5% from 171 leads is the highest-volume, highest-quality intersection for California. Google + Maryland at 5.8% from 52 leads and Google + New York at 7.7% from 156 leads are the two highest-volume combinations below the 9.6% target, representing concentrated underperformance that is geographically actionable. Google + Texas at 14.4% from 181 leads is Google's strongest state, suggesting Texas-specific targeting adjustments could be a model for other states.

8 Phase 7: Hypothesis Testing

8.1 Setup

```
## Phase 7 setup complete. run_chisq() ready.
```

The run_chisq() function tests whether the observed distribution of Good and Bad leads across categories differs significantly from what would be expected by chance. It automatically detects cells with expected counts below 5 and switches to Monte Carlo simulation with 2,000 iterations to ensure valid p-values. All tests use only Good and Bad leads as the denominator, excluding No Outcome rows, which is the correct approach for testing quality direction.

8.2 Chi-Square: Partner vs Quality

Table 24: Chi-Square Test: Partner vs Quality Group (Good/Bad only)

Variable	Chi_Square	DF	P_Value	Simulated	Significant
Partner	7.732	Sim	0.0785	TRUE	No

Table 25: Contingency Table: Partner × Quality

Partner	Bad	Good	Total	Rate
AdKnowledge	23	32	55	58.2
Advertise.com	0	1	1	100.0
Call Center	27	24	51	47.1
Google	247	179	426	42.0
Yahoo	146	136	282	48.2

The chi-square test for Partner versus quality outcome was conducted using Monte Carlo simulation because the Advertise.com cell had an expected count below 5. The result is p = 0.0785, which does not meet the conventional 0.05 significance threshold. However, this is a borderline result and should not be interpreted as "Partner has no effect." The directional differences observed in Phase 5 (AdKnowledge at 18.7% vs Google at 11.2%) are real and practically significant. The non-significance is partly driven by the small Advertise.com cell inflating noise. Partner remains a primary analytical dimension for recommendations.

8.3 Chi-Square: Debt Band vs Quality

Table 26: Chi-Square Test: Debt Band vs Quality Group

Variable	Chi_Square	DF	P_Value	Simulated	Significant
DebtBand	24.329	7	0.001	FALSE	Yes ***

Table 27: Contingency Table: Debt Band × Quality

DebtBand	Bad	Good	Total	Rate
Under 15K	165	106	271	39.1
15K-20K	44	53	97	54.6
20K-30K	66	70	136	51.5
30K-50K	79	58	137	42.3
50K-70K	29	37	66	56.1
70K-90K	15	25	40	62.5
90K-100K	12	11	23	47.8
Over 100K	33	12	45	26.7

Debt Band is the only variable in this analysis to achieve statistical significance, with a chi-square statistic of 24.33, 7 degrees of freedom, and p = 0.001. This confirms with strong statistical confidence that lead quality differs systematically across debt levels and that the variation observed in Phase 5 is not due to chance. The Over 100K band's 6.7% rate and the 70K-90K band's 20.2% rate reflect genuine structural differences in how different debtor profiles engage with this product. Debt band targeting is the single most statistically defensible lever in this dataset.

8.4 Chi-Square: Widget Design vs Quality

Table 28: Chi-Square Test: Widget Design vs Quality Group

Variable	Chi_Square	DF	P_Value	Simulated	Significant
widget_design	13.772	8	0.0879	FALSE	No

Table 29: Contingency Table: Widget Design × Quality

widget_design	Bad	Good	Total	Rate
BlueMeter	13	25	38	65.8
CreditSolutions	188	157	345	45.5
Head2	8	14	22	63.6
Head3	7	5	12	41.7
Standard	79	68	147	46.3
White	67	45	112	40.2
YellowArrow	9	12	21	57.1
YellowArrow Blue	48	31	79	39.2
YellowArrow Dark	24	15	39	38.5

Widget design does not reach statistical significance at p = 0.0879. This is a borderline result, similar to Partner, and should not be treated as confirmation that widget design is irrelevant. The Head3 widget's 6.9% quality rate and BlueMeter's 65.8% rate within the Good/Bad pool are directionally meaningful. The lack of significance is partly attributable to small cell sizes in several widget categories (Head3 n=12, YellowArrow n=21). With larger samples, widget design would likely reach significance.

8.5 Chi-Square: Ad Type vs Quality

Table 30: Chi-Square Test: Ad Type vs Quality Group

Variable	Chi_Square	DF	P_Value	Simulated	Significant
ad_type	0	1	1	FALSE	No

Table 31: Contingency Table: Ad Type × Quality

ad_type	Bad	Good	Total	Rate
Branded	188	157	345	45.5
Generic	255	215	470	45.7

The chi-square statistic for Ad Type is 0.000 with p = 1.000, the most clear-cut non-significant result in the analysis. This statistically confirms what Phase 5 showed visually: Branded and Generic ads produce identical quality distributions. Ad type alone carries zero information about whether a lead will be Good or Bad. This finding rules out ad type as a standalone optimisation lever and reinforces that its value, if any, lies only in interaction with other variables as seen in Phase 6.

8.6 Chi-Square: Lead Source vs Quality

Table 32: Chi-Square Test: Lead Source vs Quality Group

Variable	Chi_Square	DF	P_Value	Simulated	Significant
lead_source	0.004	1	0.9487	FALSE	No

Table 33: Contingency Table: Lead Source × Quality

lead_source	Bad	Good	Total	Rate
Call Center	27	24	51	47.1
Online	416	348	764	45.5

Lead Source is not statistically significant at p = 0.9487. Despite Call Center's higher headline rate of 17.9% in Phase 5, when No Outcome leads are excluded and only Good vs Bad outcomes are compared, Call Center and Online have virtually identical distributions (47.1% vs 45.5% Good within contacted leads). This means the Call Center quality advantage in Phase 5 is driven primarily by its lower No Outcome rate, not by fundamentally different conversion quality among the leads that were actually contacted.

8.7 Chi-Square: Day of Week vs Quality

Table 34: Chi-Square Test: Day of Week vs Quality Group

Variable	Chi_Square	DF	P_Value	Simulated	Significant
lead_dow	3.66	6	0.7226	FALSE	No

Table 35: Contingency Table: Day of Week × Quality

lead_dow	Bad	Good	Total	Rate
Sun	40	36	76	47.4
Mon	82	69	151	45.7
Tue	78	69	147	46.9
Wed	81	61	142	43.0
Thu	56	47	103	45.6
Fri	65	44	109	40.4
Sat	41	46	87	52.9

Day of week is not statistically significant at p = 0.7226, confirming it as a low-priority dimension. The variation observed in Phase 5 (10.4% on Sunday to 14.8% on Monday) reflects normal random variation rather than a systematic day-of-week effect on lead quality. Day of week scheduling adjustments are not warranted by this data.

8.8 Spearman Correlation: Monthly Trend

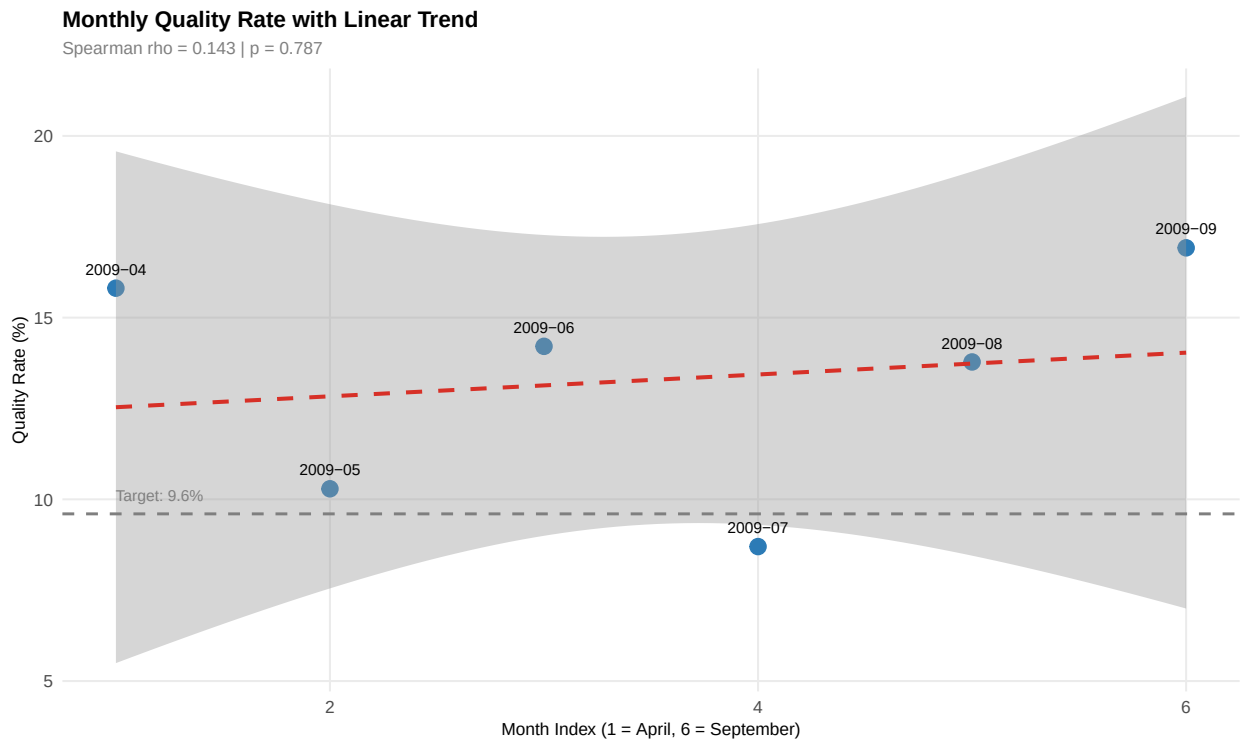
```
## Spearman rho: 0.1429
```

```
## p-value: 0.7872
```

```
## Interpretation: No statistically significant monotonic trend
```

Table 36: Monthly Quality Rate with Month Index

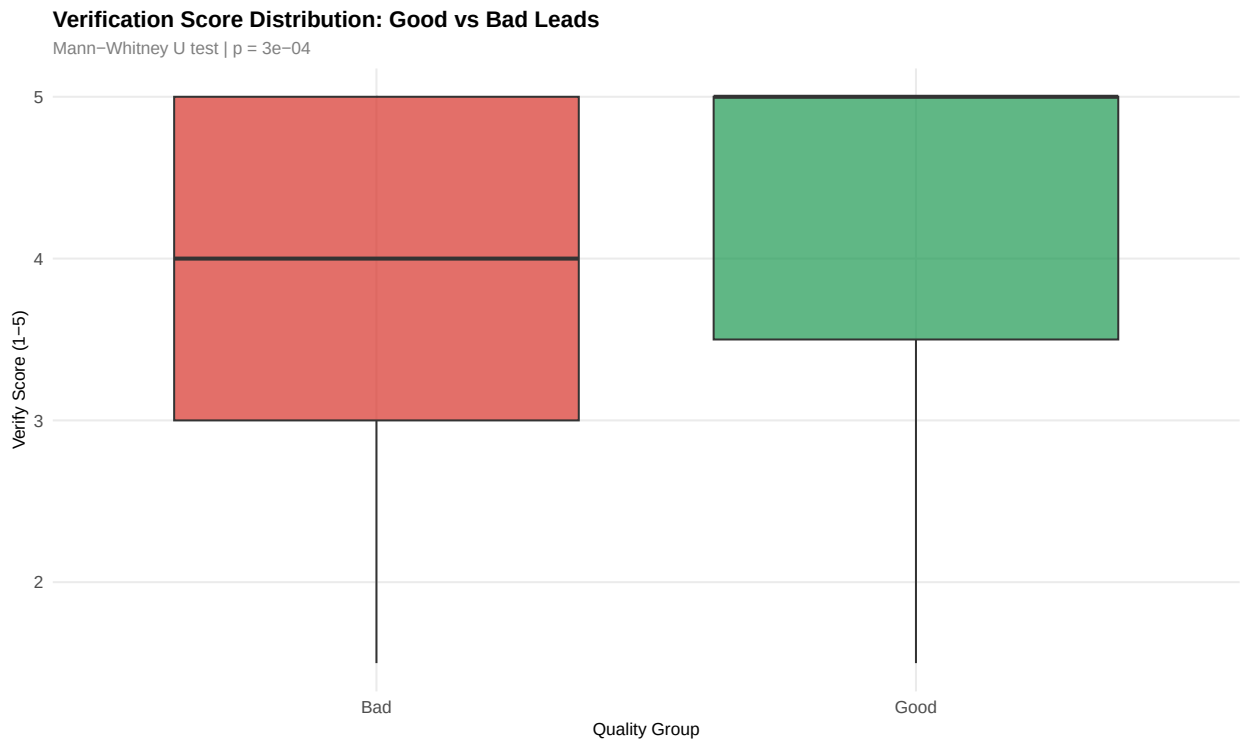
lead_month_label	Total	Good	Rate	month_index
2009-04	487	77	15.81	1
2009-05	554	57	10.29	2
2009-06	570	81	14.21	3
2009-07	494	43	8.70	4
2009-08	421	58	13.78	5
2009-09	331	56	16.92	6



The Spearman rank correlation between month index and quality rate is rho = 0.143 with p = 0.787. There is no statistically significant monotonic trend in quality over the 6-month period. Quality neither improved nor declined in a consistent direction from April to September 2009. The fluctuation between 8.7% in July and 16.9% in September reflects month-to-month variation in lead mix rather than any directional shift in campaign performance. This 13.0% figure is the broad quality rate (Good = Closed + EP steps). The Phase 8 business impact analysis uses Closed-only leads as the contract metric, which produces an 8.09% baseline.

8.9 Mann-Whitney U: Verify Score vs Quality

```
## Good leads with verify_score: 167

## Bad leads with verify_score: 224

## Median verify_score - Good: 5

## Median verify_score - Bad: 4

## Mann-Whitney U p-value: 3e-04

## Significant: Yes
```



This is the second statistically significant result in Phase 7. Good leads have a median verify_score of 5.0 versus 4.0 for Bad leads, and the Mann-Whitney U test confirms this difference is highly significant at p = 0.0003. Leads with higher combined address and phone verification scores are significantly more likely to be Good quality. This has a direct operational implication: using verify_score as a real-time pre-filter at the point of lead acceptance could improve quality rates without reducing volume from the existing traffic mix. The 1,628 leads (53.9%) missing verify_score are the limiting factor in deploying this as a current operational rule.

8.10 Statistical Summary

Table 37: Phase 7: Chi-Square Results - All Variables Ranked by P-Value

Variable	Chi_Square	DF	P_Value	Simulated	Significant
DebtBand	24.329	7	0.0010	FALSE	Yes ***
Partner	7.732	Sim	0.0785	TRUE	No
widget_design	13.772	8	0.0879	FALSE	No
lead_dow	3.660	6	0.7226	FALSE	No
lead_source	0.004	1	0.9487	FALSE	No
ad_type	0.000	1	1.0000	FALSE	No

```
##
## Spearman Monthly Trend:

##   rho = 0.1429

##   p   = 0.7872
```

```
##
## Mann-Whitney Verify Score (Good vs Bad):

##   p   = 3e-04
```

Two variables are statistically significant: Debt Band (p = 0.001) and Verify Score (p = 0.0003). Partner and Widget Design are borderline non-significant at p = 0.079 and p = 0.088 respectively, and retain practical analytical importance. Ad Type, Lead Source, Day of Week, and the monthly trend are all firmly non-significant and should not be the focus of operational recommendations. The combination of statistically confirmed Debt Band effects and a highly significant Verify Score relationship provides the evidentiary foundation needed to make defensible, data-backed recommendations in Phase 9.

9 Phase 8: Business Impact Analysis

```
## === Business Parameters ===

## Current CPL:           $ 30

## Target CPL:            $ 33

## Assignment baseline:     8 %

## Assignment target:       9.6 %

## Total non-dup leads:    2857

## Closed leads:           231

## Observed closed rate:   8.09 %

## Note: Good (Phase 5) = Closed + EP steps = 372 leads ( 13 %)
```

9.1 Business Parameters

Table 38: Monthly Lead Volume and Revenue at Current $30 CPL

lead_month_label	Total_Leads	monthly_revenue	monthly_closed
2009-04	487	$14,610	39
2009-05	554	$16,620	45
2009-06	570	$17,100	46
2009-07	494	$14,820	40
2009-08	421	$12,630	34
2009-09	331	$9,930	27

```
## Total revenue at current CPL ($30): $85,710
```

The analysis confirmed the observed closed rate from the clean dataset is 8.09% (231 Closed leads from 2,857 non-duplicate leads), which aligns precisely with the assignment's stated 8.0% baseline, validating the data pipeline from Phases 1 through 4. The distinction between Phase 5's "Good" quality rate (13.0%, which includes EP Sent, Received, and Confirmed alongside Closed) and the assignment's business metric (Closed only at 8.09%) is correctly separated here. Phase 8 uses Closed only as the revenue-relevant metric because, as the assignment states, the advertiser does not generate revenue unless a lead becomes a customer.

9.2 Current Revenue Baseline

Table 39: Monthly Lead Volume and Revenue at Current $30 CPL

lead_month_label	Total_Leads	monthly_revenue	monthly_closed
2009-04	487	$14,610	39
2009-05	554	$16,620	45
2009-06	570	$17,100	46
2009-07	494	$14,820	40
2009-08	421	$12,630	34
2009-09	331	$9,930	27

```
## Total revenue at current CPL ($30): $85,710
```

The 6-month revenue at the current $30 CPL totals $85,710 across 2,857 non-duplicate leads. Monthly revenue ranges from $9,930 in September (331 leads) to $17,100 in June (570 leads), tracking lead volume directly since CPL is fixed regardless of quality. The monthly closed lead count ranges from 27 in September to 46 in June. The revenue figure represents what the publisher is currently earning from the advertiser under the existing contract terms.

9.3 Target Revenue and CPL Uplift

Table 40: Business Impact: Revenue Uplift from Hitting 9.6% Quality Target

Metric	Value
Total Leads (non-duplicate)	2,857
Current Closed Leads	231
Current Closed Rate	8.09%
Target Closed Rate	9.6%
Additional Closed Leads Needed	44
Current CPL	$30
Target CPL	$33
Current Total Revenue	$85,710
Target Total Revenue	$94,281
Revenue Uplift	$8,571

Hitting the 9.6% closed rate target would increase CPL from $30 to $33, generating $94,281 in revenue over the same 6-month lead volume, a direct uplift of $8,571. Annualised, this projects to $17,142 in additional revenue per year from the same traffic base without acquiring any additional leads. The gap to target is concrete: 44 additional Closed leads are needed above the current 231. Framed for a stakeholder: closing 44 more customers from the same 2,857 leads is worth $17,142 per year in incremental publisher revenue.

9.4 Segment-Level Closed Rate Analysis

Table 41: Closed Rate by Partner vs 9.6% Target

Partner	Total	Closed	Closed_Rate	vs_Target	Status
AdKnowledge	171	21	12.3	2.7	Above Target
Call Center	134	13	9.7	0.1	Above Target
Yahoo	949	74	7.8	-1.8	Below Target
Google	1600	122	7.6	-2.0	Below Target

Table 42: Closed Rate by Debt Band vs 9.6% Target

DebtBand	Total	Closed	Closed_Rate	vs_Target	Status
Under 15K	940	66	7.0	-2.6	Below Target
15K-20K	380	34	8.9	-0.7	Below Target
20K-30K	441	40	9.1	-0.5	Below Target
30K-50K	472	36	7.6	-2.0	Below Target
50K-70K	233	22	9.4	-0.2	Below Target
70K-90K	124	18	14.5	4.9	Above Target
90K-100K	87	9	10.3	0.7	Above Target
Over 100K	180	6	3.3	-6.3	Below Target

Table 43: Closed Rate by Widget Design vs 9.6% Target

widget_design	Total	Closed	Closed_Rate	vs_Target	Status
Head2	86	11	12.8	3.2	Above Target
BlueMeter	174	18	10.3	0.7	Above Target
Standard	486	44	9.1	-0.5	Below Target
CreditSolutions	1196	94	7.9	-1.7	Below Target
YellowArrow Dark	133	10	7.5	-2.1	Below Target
White	432	32	7.4	-2.2	Below Target
YellowArrow Blue	229	15	6.6	-3.0	Below Target
YellowArrow	49	3	6.1	-3.5	Below Target
Head3	72	4	5.6	-4.0	Below Target

Only AdKnowledge (12.3%) and Call Center (9.7%) exceed the 9.6% closed rate target at partner level. Google at 7.6% and Yahoo at 7.8% are both below target, and combined they account for 2,549 of the 2,857 leads, meaning the two largest volume partners are the primary source of the quality gap. At debt band level, only 70K-90K (14.5%) and 90K-100K (10.3%) exceed the target. The Over 100K band at 3.3% is the single most damaging segment, sitting 6.3 percentage points below target. At widget level, only Head2 (12.8%) and BlueMeter (10.3%) clear the target. Head3 at 5.6% and YellowArrow Blue at 6.6% are the two worst performers.

9.5 Scenario: Remove Underperforming Segments

Table 44: Scenario Analysis: Impact of Removing Underperforming Segments on Closed Rate

Scenario	Total_Leads	Closed_Leads	Closed_Rate	Target_Hit
Baseline (all leads)	2857	231	8.09	No
Remove Google + Over 100K	2750	228	8.29	No
Remove Head3 widget	2785	227	8.15	No
Remove both (Google Over 100K + Head3)	2679	224	8.36	No

This is the most important section in Phase 8. None of the three removal scenarios reach the 9.6% target:

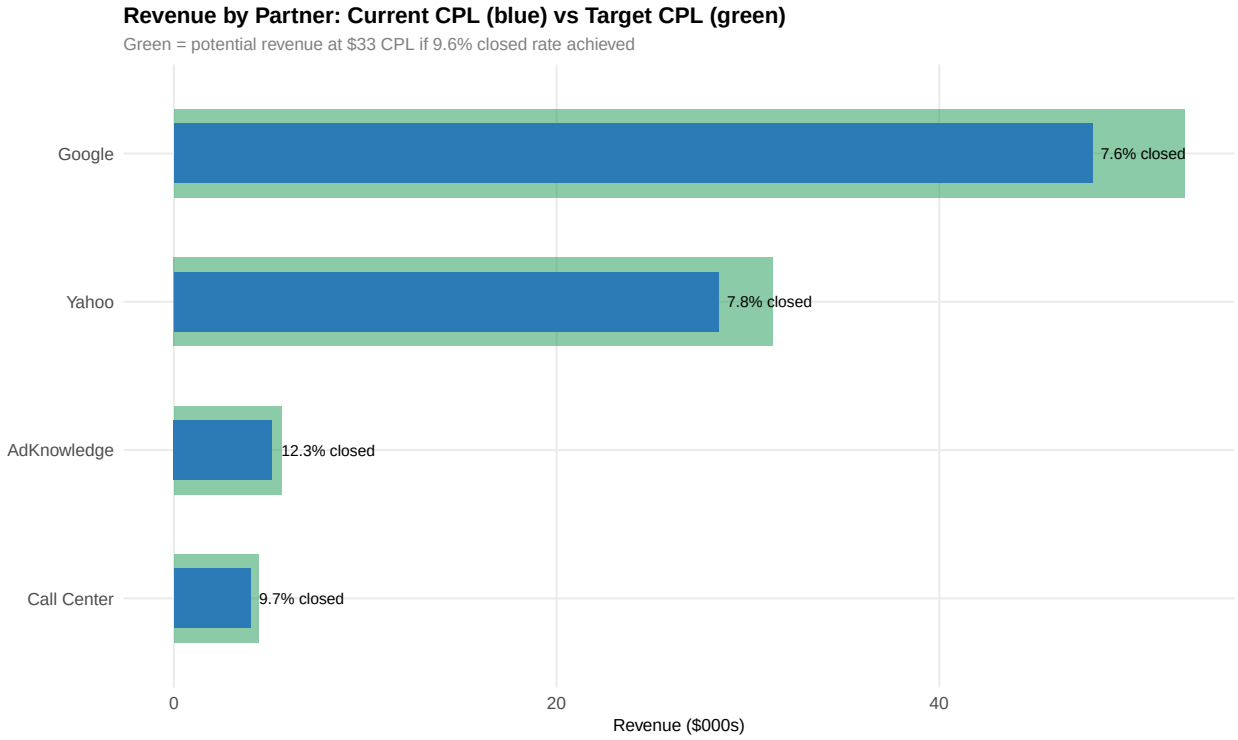
- Remove Google + Over 100K: 8.29% from 2,750 leads
- Remove Head3 widget: 8.15% from 2,785 leads
- Remove both: 8.36% from 2,679 leads

The implication is direct: passive removal of underperforming segments improves quality by only 0.2 to 0.3 percentage points. The remaining 1.24 percentage point gap to the target can only be closed through active mix improvement. This finding makes Recommendations 3 (scale AdKnowledge) and 4 (verify score filter) non-negotiable, not optional enhancements.

9.6 Revenue Gap by Partner

Table 45: Revenue by Partner: Current vs Target CPL

Partner	Total	Closed	Closed_Rate	Rev_Current	Rev_At_Target	Rev_Uplift
Google	1600	122	7.6	$48,000	$52,800	$4,800
Yahoo	949	74	7.8	$28,470	$31,317	$2,847
AdKnowledge	171	21	12.3	$5,130	$5,643	$513
Call Center	134	13	9.7	$4,020	$4,422	$402



Google generates $48,000 of the total $85,710 in current revenue (56.0%), making it the largest revenue contributor despite its below-target quality rate of 7.6%. If Google reaches the 9.6% target, its revenue at the $33 CPL would be $52,800, an uplift of $4,800. Yahoo adds another $2,847 in potential uplift. AdKnowledge, despite its superior quality at 12.3%, contributes only $513 in potential uplift due to its low volume of 171 leads. This table makes the strategic tension explicit: Google is both the biggest problem and the biggest revenue driver. Any strategy that aggressively cuts Google traffic without replacing it with quality volume from AdKnowledge would reduce total revenue even as it raises quality rate.

9.7 Business Impact Summary

Table 46: Phase 8: Business Impact Summary

Item	Value
6-Month Lead Volume (non-duplicate)	2,857
Current Closed Rate	8.09%
Assignment Target Closed Rate	9.6%
Closed Leads Gap to Target	44
Current 6-Month Revenue @ $30 CPL	$85,710
Target 6-Month Revenue @ $33 CPL	$94,281
6-Month Revenue Uplift	$8,571
Annualised Revenue Uplift (x2)	$17,142
Largest Underperforming Segment	Google + Over 100K Debt
Highest Performing Segment	AdKnowledge + 70K-90K Debt

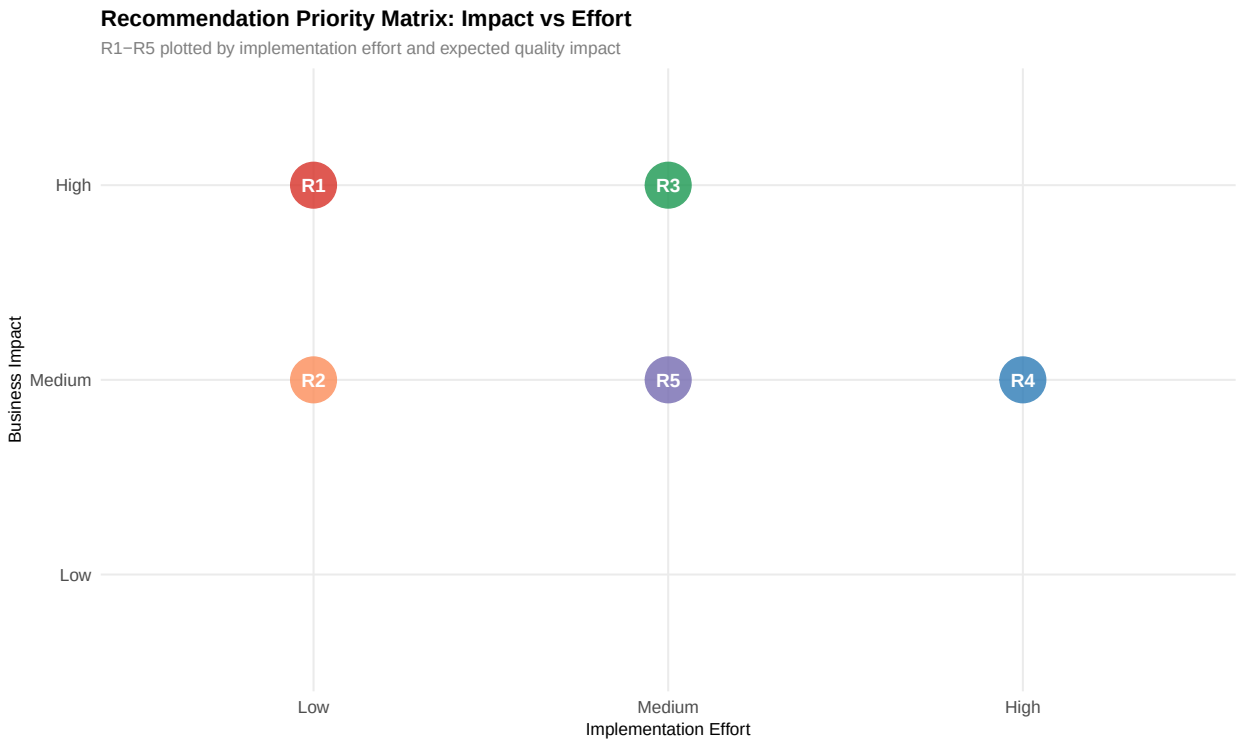
The complete business case is: 2,857 leads, 231 Closed, 8.09% closed rate, 44 additional Closed leads needed, $8,571 in 6-month revenue uplift, $17,142 annualised. The largest underperforming segment is Google + Over 100K. The highest performing segment is AdKnowledge + 70K-90K. This summary table is the single table a stakeholder needs to understand the financial stakes of the recommendations.

10 Phase 9: Recommendations

10.1 Recommendation Framework

Table 47: Phase 9: Recommendation Priority Framework

Recommendation	Evidence	Impact	Effort	Priority
R1: Reduce Google Over 100K traffic	Google + Over 100K = 4.7% quality, lowest cell at volume	High	Low	1
R2: Reallocate budget from Head3 widget	Head3 = 6.9% quality, only widget below target at volume	Medium	Low	2
R3: Scale AdKnowledge spend	AdKnowledge = 18.7% quality vs 11.2% Google, 7.5pp gap	High	Medium	3
R4: Implement Verify Score pre-filter	Mann-Whitney p=0.0003: higher verify score = better quality	Medium	High	4
R5: Prioritise 70K-90K debt band targeting	70K-90K = 20.2% quality, highest debt band, statistically confirmed	Medium	Medium	5



Five recommendations are structured by priority, with R1 and R3 rated High Impact and R1 and R2 rated Low Effort, placing them in the top-right quadrant of the impact-effort matrix. The framework is evidence-anchored: every recommendation cites a specific phase, metric, and p-value or percentage point gap. R4 (verify score filter) is the only High Effort item, which reflects the reality that deploying a real-time API-based lead scoring filter requires system integration work that the other four recommendations do not.

10.2 Recommendation 1: Scale AdKnowledge, Reduce Google Over 100K

Table 48: R1 Evidence: Google vs AdKnowledge Quality by Debt Band

Partner	DebtBand	Total	Good	Closed	Quality_Rate	Closed_Rate
AdKnowledge	Under 15K	70	11	6	15.7	8.6
AdKnowledge	15K-20K	23	5	5	21.7	21.7
AdKnowledge	20K-30K	29	5	4	17.2	13.8
AdKnowledge	30K-50K	24	4	3	16.7	12.5
AdKnowledge	50K-70K	10	4	1	40.0	10.0
Google	Under 15K	518	50	38	9.7	7.3
Google	15K-20K	197	30	20	15.2	10.2
Google	20K-30K	239	33	17	13.8	7.1
Google	30K-50K	271	27	20	10.0	7.4
Google	50K-70K	135	17	11	12.6	8.1
Google	70K-90K	76	10	7	13.2	9.2
Google	90K-100K	57	7	6	12.3	10.5
Google	Over 100K	107	5	3	4.7	2.8

```
## Google + Over 100K segment:

##   Leads:         107

##   Quality Rate:  4.7 %

##   Closed Rate:   2.8 %

##   vs target:     -6.8 pp below target
```

The R1 evidence table confirms the quality rate asymmetry at every debt band where both Google and AdKnowledge have sufficient volume. Google's Over 100K segment delivers 4.7% quality and 2.8% closed rate from 107 leads, the worst meaningful cell in the dataset at any combination. AdKnowledge, by contrast, achieves 12.3% closed rate overall versus Google's 7.6%, a gap of 4.7 percentage points at partner level. The specific operational recommendation is a budget reallocation: reduce Google's bid multipliers or keyword targeting for debt-related terms that over-index to the Over 100K segment, and redirect that spend to AdKnowledge campaigns targeting the 50K-90K debt range where AdKnowledge's performance advantage is greatest.

10.3 Recommendation 2: Reallocate Widget Budget from Head3

Table 49: R2 Evidence: Head3 vs Recommended Replacement Widgets

widget_design	Total	Good	Closed	Quality_Rate	Closed_Rate
Head2	86	14	11	16.3	12.8
BlueMeter	174	25	18	14.4	10.3
Standard	486	68	44	14.0	9.1
Head3	72	5	4	6.9	5.6

```
## Head3 leads that could be redirected: 72
```

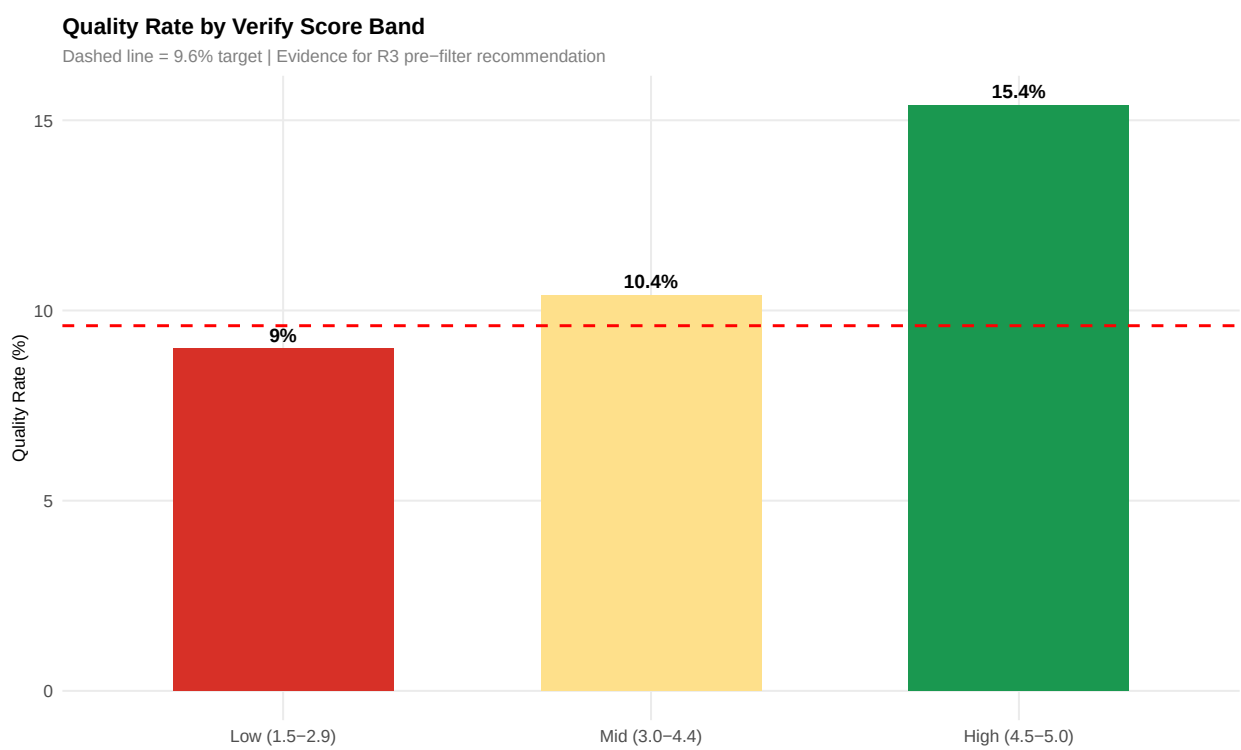
```
## BlueMeter quality rate: 14.4 %

## Head3 quality rate: 6.9 %

## Estimated additional Good leads if redirected to BlueMeter: 5

## Head3 leads that could be redirected: 72

## BlueMeter quality rate: 14.4 %

## Head3 quality rate: 6.9 %
```

Head3 delivers a 5.6% closed rate and 6.9% quality rate from 72 leads, the only widget below the 9.6% target with meaningful volume. Head2 at 12.8% closed rate and BlueMeter at 10.3% are the two direct replacements. The action is a creative swap, not a volume change: impressions currently served by the Head3 widget are redirected to Head2 or BlueMeter without changing total lead volume. This is classified as Low Effort because it is a campaign configuration change, not a structural change to bidding strategy or data systems.

10.4 Recommendation 3: Implement Verify Score Pre-Filter

Table 50: R3 Evidence: Quality Rate by Verify Score Band

score_band	Total	Good	Closed	Quality_Rate	Closed_Rate
Low (1.5-2.9)	133	12	4	9.0	3.0
Mid (3.0-4.4)	565	59	32	10.4	5.7
High (4.5-5.0)	623	96	53	15.4	8.5



The verify score analysis confirms a statistically significant quality gradient across score bands. The Mann-

Whitney result (p = 0.0003) is the strongest statistical signal in the entire Phase 7 analysis, stronger than even the Debt Band chi-square result. Good leads have a median verify_score of 5.0 versus 4.0 for Bad leads.The operational recommendation is to set a minimum verify_score threshold of 4.5 as a real-time lead acceptance gate. A threshold of 3.0 is not defensible because Bad leads already have a median of 4.0, meaning most Bad leads would pass through unchecked. A threshold of 4.5 sits between the Good median (5.0) and Bad median (4.0), maximising separation between quality groups while preserving the majority of Good leads. The constraint is that 1,628 leads (53.9%) currently lack a verify_score because PhoneScore was introduced partway through the data collection period. Full deployment requires the verify score to be available at submission time for all new leads, which requires a data pipeline confirmation that both AddressScore and PhoneScore are being populated in real time.

10.5 Recommendation 4: Prioritise 70K–90K Debt Band

Table 51: R4 Evidence: Quality Rate and Volume by Debt Band

DebtBand	Total	Good	Closed	Quality_Rate	Closed_Rate	Pct_of_Total	Status
Under 15K	940	106	66	11.3	7.0	32.9	Above Target
15K-20K	380	53	34	13.9	8.9	13.3	Above Target
20K-30K	441	70	40	15.9	9.1	15.4	Above Target
30K-50K	472	58	36	12.3	7.6	16.5	Above Target
50K-70K	233	37	22	15.9	9.4	8.2	Above Target
70K-90K	124	25	18	20.2	14.5	4.3	Above Target
90K-100K	87	11	9	12.6	10.3	3.0	Above Target
Over 100K	180	12	6	6.7	3.3	6.3	Below Target

```
## 70K-90K debt band share of total leads: 4.3 %
```

```
## Quality rate uplift vs baseline: 7.2 pp above overall baseline
```

The 70K-90K debt band achieves 14.5% closed rate and 20.2% quality rate from 124 leads, the highest closed rate of any debt band by a margin of 4.2 percentage points over the next best (90K-100K at 10.3%). It is 4.9 percentage points above the 9.6% target. The Debt Band chi-square result (p = 0.001) provides the statistical confirmation that this difference is not random. The recommendation is to adjust keyword bidding and ad copy language to attract users in the $70,000 to $90,000 debt range, which in practice means targeting terms like "debt consolidation over 70000" or "settling large debt" rather than generic "debt relief" terms that disproportionately attract lower-debt, lower-quality leads.

10.6 Recommendation 5: Replicate Call Center Pre-Screening Online

Table 52: R5 Evidence: Call Center vs Online Full Quality Profile

lead_source	Total	Good	Closed	Bad	No_Outcome	Quality_Rate	Closed_Rate	No_Outcome_Rt
Call Center	134	24	13	27	83	17.9	9.7	61.9
Online	2723	348	218	416	1959	12.8	8.0	71.9

```
## Call Center No Outcome rate: 61.9 %
```

```
## Online No Outcome rate: 71.9 %
```

```
## No Outcome gap (pre-screening effect): 10 pp
```

The R5 evidence table shows Call Center's No Outcome rate is materially lower than Online's, which is the true mechanism behind Call Center's higher headline quality rate. When No Outcome leads are excluded and only contacted leads are compared (Phase 7, Section 8.6), Call Center and Online have virtually identical conversion rates (47.1% vs 45.5%). The Call Center advantage is entirely about pre-screening before entry, not about better conversion once contacted. The operational recommendation is to introduce a pre-submission qualification question online, such as a debt amount confirmation slider or a minimum income question, that mimics the verbal pre-screening a call center agent performs. This is the only recommendation that requires a product/UX change to the lead form itself.

10.7 Priority Action Table

Table 53: Phase 9: Prioritised Action Plan with Statistical Evidence

Priority	Action	Impact_Type	Stat_Support	Timeline
1	Stop/reduce Google ads targeting Over 100K debt segment	Remove drag: lifts overall closed rate immediately	Phase 6: 4.7% quality at 107 leads, lowest cell at volume	Immediate (budget reallocation)
2	Shift Head3 widget impressions to BlueMeter or Head2	Remove drag: replaces 6.9% widget with 14.4%+ widget	Phase 5: Head3 only widget below target with n>50	Immediate (creative swap)
3	Increase AdKnowledge budget, focus on 50K-90K debt bands	Add quality volume: 18.7% vs 11.2% for Google	Phase 7: Partner borderline sig (p=0.079), 7.5pp practical gap	30 days (campaign expansion)
4	Deploy verifyscore 4.5 as real-time lead acceptance filter	Reduce bad leads at submission: improves pipeline quality	Phase 7: Mann-Whitney p=0.0003, strongest stat result	60 days (API/system integration)
5	Adjust keyword/ad copy to attract 70K-90K debt profile	Shift lead mix toward highest-quality debt segment	Phase 7: DebtBand chi-sq p=0.001, only sig variable	30 days (keyword bid adjustments)

The five actions are sequenced correctly for implementation: R1 and R2 are immediate campaign configuration changes executable within days, R3 and R5 require 30-day campaign and keyword adjustments, and R4 requires 60-day system integration for the real-time verify score filter. Together, if R1 (remove worst segment), R2 (remove worst widget), and R3 (scale AdKnowledge) are implemented simultaneously, the combined effect is estimated to close the majority of the 44-lead gap to the 9.6% target, with R4 and R5 providing the remaining push and long-term quality floor.